

## 41005 Series

Cage Guided Globe Valve with Lo-dB  
Anti-cavitation and API 6A Capabilities

Instruction Manual (Rev. H)



THESE INSTRUCTIONS PROVIDE THE CUSTOMER/OPERATOR WITH IMPORTANT PROJECT-SPECIFIC REFERENCE INFORMATION IN ADDITION TO THE CUSTOMER/OPERATOR'S NORMAL OPERATION AND MAINTENANCE PROCEDURES. SINCE OPERATION AND MAINTENANCE PHILOSOPHIES VARY, BAKER HUGHES COMPANY (AND ITS SUBSIDIARIES AND AFFILIATES) DOES NOT ATTEMPT TO DICTATE SPECIFIC PROCEDURES, BUT TO PROVIDE BASIC LIMITATIONS AND REQUIREMENTS CREATED BY THE TYPE OF EQUIPMENT PROVIDED.

THESE INSTRUCTIONS ASSUME THAT OPERATORS ALREADY HAVE A GENERAL UNDERSTANDING OF THE REQUIREMENTS FOR SAFE OPERATION OF MECHANICAL AND ELECTRICAL EQUIPMENT IN POTENTIALLY HAZARDOUS ENVIRONMENTS. THEREFORE, THESE INSTRUCTIONS SHOULD BE INTERPRETED AND APPLIED IN CONJUNCTION WITH THE SAFETY RULES AND REGULATIONS APPLICABLE AT THE SITE AND THE PARTICULAR REQUIREMENTS FOR OPERATION OF OTHER EQUIPMENT AT THE SITE.

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## Safety Information

### Important - Please Read Before Installation

Masoneilan™ 41005 Series instructions contain **DANGER**, **WARNING**, and **CAUTION** labels, where necessary, to alert you to safety related or other important information. Read the instructions carefully **before** installing and maintaining your control valve. **DANGER** and **WARNING** hazards are related to personal injury. **CAUTION** hazards involve equipment or property damage. Operation of damaged equipment can, under certain operational conditions, result in degraded process system performance that can lead to injury or death. Total compliance with all **DANGER**, **WARNING**, and **CAUTION** notices is required for safe operation.



This is the safety alert symbol. It alerts you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.



Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



Indicates a potentially hazardous situation which, if not avoided, could result in serious injury.



Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury.



When used without the safety alert symbol indicates a potentially hazardous situation which, if not avoided, could result in property damage.

**Note:** Indicates important facts and conditions.

## About this Manual

- The information in this manual is subject to change without prior notice.
- The information contained in this manual, in whole or part, shall not be transcribed or copied without Baker Hughes written permission.
- Please report any errors or questions about the information in this manual to your local supplier.
- These instructions are written specifically for the 41005 Series control valves, and do not apply for other valves outside of this product line.

### Useful Life Period

The current estimated useful life period for the Masoneilan 41005 Series control valve is 25+ years. To maximize the useful life of the product it is essential to conduct annual inspections, routine maintenance and ensure proper installation to avoid any unintended stresses on the product. The specific operating conditions will also impact the useful life of the product. Consult the factory for guidance on specific applications if required prior to installation.

### Warranty

Items sold by Baker Hughes are warranted to be free from defects in materials and workmanship for a period of one year from the date of shipment provided said items are used according to Baker Hughes recommended usages. Baker Hughes reserves the right to discontinue manufacture of any product or change product materials, design or specifications without notice.

**Note:** Prior to installation:

- The valve must be installed, put into service and maintained by qualified and competent professionals who have undergone suitable training.
- Under certain operating conditions, the use of damaged equipment could cause a degradation of the performance of the system which may lead to personal injury or death.
- Changes to specifications, structure, and components used may not lead to the revision of this manual unless such changes affect the function and performance of the product.
- All surrounding pipe lines must be thoroughly flushed to ensure all entrained debris has been removed from the system.

# WARNING

## Safety Information (cont.)

1. It is the end user's responsibility to ensure properly loaded and supported piping to avoid undesired stresses on the product which may result in damage to the product, loss of containment, or loss of functionality and resulting unsafe states or conditions.
2. It is the end user's responsibility to correctly identify end locations and place product in areas that may contain explosive atmospheres. The failure of properly following the test, installation, maintenance and disassembly and assembly instructions may result in a compromised product which in turn could result in an uncontrolled or unexpected loss of containment and release of pressure.
3. It is the end user's responsibility to take appropriate actions to ensure that site personnel who are performing installation, commissioning, and maintenance have been trained in proper site procedures for working with and around Baker Hughes supplied equipment, per Safe Site Work Practices.
4. The failure of properly following the test, installation, maintenance, disassembly and assembly instructions may result in a compromised product which in turn could result in an uncontrolled or unexpected loss of containment and release of pressure. It is the responsibility of the person conducting the tasks listed above to take great care in following such procedures.
5. It is the end user's responsibility to:
  - Recognize and safely contain any leak.
  - Ensure proper Personal Protective Equipment is available and used.
  - Follow Proper lifting techniques, equipment and procedures, per Safe Site Work Practices.
6. Proper Lockout and Tagout of energy sources prior to maintenance, per Safe Site Work Practices, is the end user's responsibility. This includes any potential control signals or circuits that may have a remote or automated control function over any product. Instructions to properly release spring stored energy are included in this manual.
7. After installation or maintenance, it is the end user's responsibility to ensure the equipment has been properly inspected and returned to proper condition before being returned to service.

# 1. General

## Important

This document contains all the instructions required for the installation, operation and maintenance of the equipment. Regular maintenance, strict observance of these instructions and the use of manufacturer's replacement parts will guarantee optimum operation and reduce maintenance costs.

### 1.1 Scope

The following instructions are designed to guide the user through the installation and maintenance of the 41005 Series valves for all sizes and all pressure classes.

### 1.2 Serial Plate

The serial plate is usually fixed to the side of the actuator yoke. It indicates, amongst other things, the type of valve, the pressure class, the material used for the pressurized chamber and the air supply pressure of the actuator.

### 1.3 After Sales Service

Baker Hughes offers its customers an After Sales Service comprising highly qualified technicians for the operation, maintenance and repair of its equipment. To benefit from this service, contact your local representative or the After Sales Service whose email address is given at the end of this document.

### 1.4 Spare Parts

When carrying out maintenance operations, only manufacturer's replacement parts must be used, obtained through our local representatives or our Spare Parts Service.

When ordering spare parts, the model and serial numbers indicated on the manufacturer's serial plate must be given.

The recommended spare parts are indicated in the parts list included in this instruction manual on page 35.

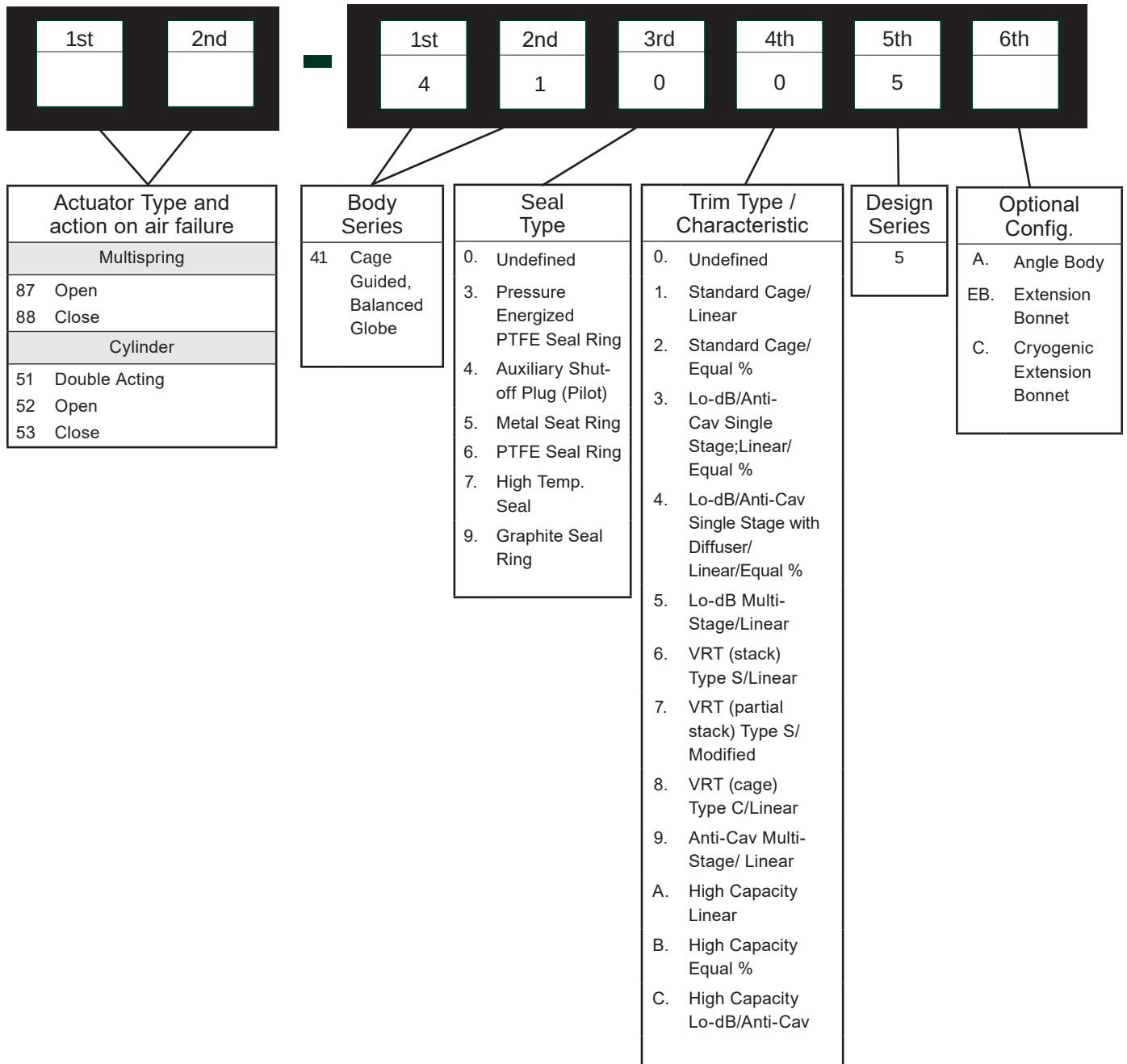
### 1.5 Actuator and Other Accessories

The valve is equipped with an actuator. Like other valve accessories, actuators have specific instructions regarding the installation, operation and maintenance for the electric and pneumatic connections.

For additional information, please refer to the instruction, operation and maintenance manuals for the 51/52/53 Series Pneumatic Piston Actuators and the 87/88 Series Actuators.



## 2. Numbering System



**Note:** This instruction manual describes all the standard options in the 41005 Series valve. To satisfy the particular conditions of your application, Baker Hughes may have had to develop a special option which is the subject of an additional clause to this instruction manual. In this case, the instructions given in the additional clause always take precedence over the general instructions.

## 3. Installation

### 3.1 Cleanness of Piping

Before installing the valve in the line, clean piping and valve of all foreign material such as welding chips, scale, oil, grease or dirt. Gasket surfaces must be thoroughly cleaned to insure leak-free joints.

### 3.2 Isolating By-Pass Valve

To allow for in-line inspection, maintenance and removal of the valve without service interruption, provide a manually operated stop valve on each side of the control valve and a manually operated throttling valve in the by-pass line.

### 3.3 Heat Insulation

In case of heat insulated installation, do not insulate the valve bonnet and take protection measures related to personal safety.

### 3.4 Hydraulic Test and Cleaning of Lines

During this operation, the control valve must not be used as an isolating valve.

This means that the valve must always be opened before carrying out pressure tests in the process line, cleaning of pipes, etc., otherwise equipment damages or destroying of seal rings could result.

### 3.5 Flow Direction

The valve must be installed so that the controlled substance will flow through the valve in the direction indicated by the flow arrow located on the body.

## 4. Disassembly

### 4.1 Removal of Actuator

Access to the internal components of the body should be accomplished with the actuator removed. To carry out this operation, follow the instructions in the Installation, Operation and Maintenance manuals for actuator types 87/88 and 51/52/53.

#### 4.1.1 Disconnection of instruments

Disconnect all mechanical couplings between the positioner and the other instruments on one side and the valve stem/ actuator stem coupling on the other side.

#### 4.1.2 Plug stems screwed into the actuator stem

In the case of air-to-retract actuators, apply sufficient air pressure on the diaphragm to retract the stem completely. Loosen the counter-nut, unscrew the stem.

### CAUTION

During this operation, make sure that the plug does not turn when it is seated. If the plug travel is very small and there is a large amount of plug stem inside the actuator, it may be necessary to remove the yoke nut and lift the actuator so that the plug is not touching the seat.

#### 4.1.3 Stems attached with a stem connector

In the case of air-to-retract actuators, apply sufficient air pressure on the diaphragm to retract the stem completely. Loosen the screws and remove the stem connector.

#### 4.1.4 Removal of actuator

Disconnect all the ingoing and outgoing air and electrical connections from the actuator. Loosen the yoke nut or attachment screws and lift the actuator, making sure that the concentricity and the thread of the bonnet is not damaged.

### 4.2 Opening of the Pressurized Chamber

(Figures 7, 8, 9, 10 and 11 on page 24)

### ! DANGER

Prior to disassembly, vent the process pressure and isolate the valve if necessary.

**Note: The valve must always be reassembled with new packing rings and gaskets. Before disassembly, make sure that the appropriate parts are available.**

- A. Remove the packing flange nuts (3) then remove the packing flange (4) and the packing follower (23).

- B. Check that the exposed part of the valve plug stem (1) is clean enough for the bonnet (7) to be removed easily.
- C. Remove the body stud nuts (8).
- D. By means of a pad eye secured instead of the actuator, lift the bonnet (7) up and separate it from the valve body (18). During this operation, the valve plug stem (1) must be pushed downwards so that the valve plug remains in the valve body (18).
- E. Remove the spring washer (17) and the body gasket (10).
- F. In the case of type 41305, 41375, 41505, 41605 and 41905 valves, remove the valve plug stem (1) and valve plug (15) assembly from the cage by pulling the valve plug stem upward, then remove the cage (16) and 'stack' (37) assembly for 41375 type valve.

In case of a 41905 series valve [equipped with a graphite ring (45)], make sure that the ring is not damaged during the operation.

### CAUTION

Because of the plug balance seal (31), the cage can sometimes be lifted along with the valve plug. If this should happen, press down on the cage so that it remains in the body. If the cage is lifted along with the valve plug, it could slip during handling and fall.

- G. In case of a 41405 valve, remove the valve plug and body cage assembly by pulling the valve stem upward: in this case, the valve plug has a shoulder which prevents the cage from falling. Remove the valve plug from the cage by pulling on the end of the plug stem.
- H. Remove the seat ring (13) and the seat ring gasket (14) from the valve body (18).
- I. Remove the packing (6), the packing spacer (5) and the guide bushing (22) from the bonnet (7).

**Note: A packing spacer (5) is only mounted when the bonnet has a side connection.**

### 4.3 Disassembly of Valve Plug Stem

The valve plug stem is screwed and pinned into the valve plug (15).

To dismount the stem, the valve plug must be held as indicated below, taking care not to damage the guiding surfaces; the plug stem pin (9 ) is then removed. By means of flats or using a nut and counter-nut on the end of the stem, unscrew the stem taking care not to apply a bending moment which could deform it.

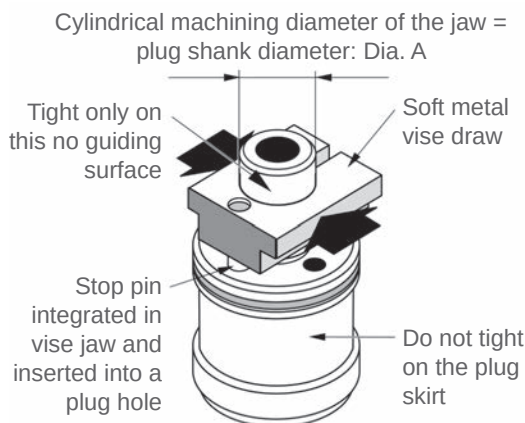


Figure 1 - Valve Plug Stem

### 4.4 Disassembly of the Auxiliary Pilot Plug

For valve type 41405 size 2" to 4" (50 to 100 mm) - see Figure 7 page 24.

Exert a sufficient force on the auxiliary pilot plug (20) to compress the spring washers (12). The retaining ring (19) can then be removed, releasing the auxiliary pilot plug and spring washers.

For valve type 41405 size 6" to 16" (150 to 400 mm) - see Figure 8 on page 24.

To carry out this operation safely, screws of the diameter and length indicated in the table below must be used. Thread the socket head cap screws through the holes in the auxiliary pilot plug (20). Tighten until the retainer ring (19) can be removed. Loosen the screws gradually. Remove the auxiliary pilot plug and the spring (12).

| Valve size |     | Pilot Dismounting Screws Size |        |      |                  |
|------------|-----|-------------------------------|--------|------|------------------|
| mm         | in. | Qty.                          | Length |      | Dia.             |
|            |     |                               | mm     | in.  |                  |
| 150        | 6   | 2                             | 57     | 2.25 | 1/4" – 20 UNC 2A |
| 200        | 8   | 2                             | 70     | 2.75 |                  |
| 250        | 10  | 2                             | 63.5   | 2.5  |                  |
| 300        | 12  | 3                             | 101.5  | 4    |                  |
| 400        | 16  | 3                             | 63.5   | 2.5  |                  |

### 4.5. Disassembly of 41705 HTS Plug

This plug type can only be serviced in the field by factory qualified Masoneilan valve maintenance technicians due to critical features that cannot be easily addressed in the field.

**Note to factory qualified Masoneilan valve technicians:**  
See the latest revision of CES-189 for instructions and maintenance values.

The metal piston rings (11) can be replaced in the field per instructions for plug type 41505.

## 5. Maintenance – Repair

### 5.1 Packing Box

Packing box maintenance is one of the principle action items of routine servicing. Tightness of the packing is maintained by packing compression. Compression is achieved by evenly tightening the packing flange nuts (3) against the packing flange (4). Care must be taken not to overtighten as this could prevent smooth operation of the valve. If packing is fully compressed and valve leaks, then new packing is required.

## CAUTION

Valve must be isolated and the pressure vented before performing packing box maintenance.

#### 5.1.1 Braided PTFE with Carbon or Aramid Core (Standard)

**Note: The Braided PTFE/Carbon or Aramid packing rings have a skive cut allowing packing replacement without disconnecting the plug stem from the actuator connector or actuator stem.**

- Loosen and remove packing flange nuts (3).
- Lift the packing flange (4) and packing follower (23) up the valve stem.

**Note: They may be taped in place to keep them out of the way before proceeding.**

- By means of a hooked instrument, remove packing rings ensuring not to damage the sealing surfaces of the packing box or plug stem.

**Note: On valves equipped with an optional lubricator connection, the lantern ring must also be removed to gain access to lower packing rings.**

- Replace packing rings.

**Note: Assemble and compress rings one at a time into the packing box. The skive cut of each packing ring must be placed about 120 degrees apart.**

**Note: On valves equipped with an optional lubricator connection, pay close attention to the number and order of packing rings and how they are arranged in the packing box. This will help with assembly.**

- E. Replace packing follower (23) and packing flange (4).
- F. Replace and tighten packing stud nuts (3).

## CAUTION

Do not overtighten.

- G. Place valve back in service and only tighten packing as necessary to stop external leakage.

**Note:** *In an emergency, string packing may be used as a temporary repair only. It must be replaced with the correct packing as soon as possible.*

### 5.1.2 Flexible Graphite Rings

**Note:** *Flexible graphite packing rings replacement may require disconnecting the plug stem from the actuator stem and removal of the actuator if rings are not skive cut.*

- A. Loosen and remove packing flange nuts (3).
- B. Remove packing flange (4) and packing follower (23) from the plug stem.
- C. By means of a hooked instrument, remove packing (6) ensuring not to damage the sealing surfaces of the packing box or plug stem.

**Note:** *On valves equipped with an optional lubricator connection, the lantern ring must also be removed to gain access to lower packing rings.*

- D. Replace new packing set (6); first assemble a back-up ring (*Graphite Filament Yarn braided ring*), then the flexible graphite rings (*smooth rings*), and finally another braided back-up ring.

**Note:** *Assemble and compress rings one at a time into packing box.*

**Note:** *On valves equipped with an optional lubricator connection, pay close attention to the number and order of packing rings and how they are arranged in the packing box. This will help with assembly.*

- E. Assemble packing follower (23) and packing flange (4).
- F. Assemble and tighten packing stud nuts (3).

## CAUTION

Do not overtighten.

- G. Proceed to appropriate instructions for actuator and valve assembly adjustment.
- H. Place valve back in service and only tighten packing as necessary to stop external leakage.

### 5.1.3 Low Emissions Packing

The Masoneilan "Low-E" (Low Emissions) Packing from Baker Hughes is a high performance packing system capable of containing fugitive emissions well below the specifications of the most severe recommendations. It is also available in a fire proof configuration if required.

The packing is provided as a set of rings consisting of middle rings and bound by end rings also referred to as anti-extrusion rings. All of our low emission solutions come with live loading

which is imperative to maintain a constant load on the packing and is necessary for thermal cycling applications.

Assembled properly, this packing provides better performance than current regulation requirements. Consequently, it can effectively prevent fugitive emissions leaks from a control valve. The "Low-E" packing system can directly replace conventional packing, requiring no modification to the control valve or actuator.

The packing material could vary depending on the specifications and timing when the valve was ordered. It is important to understand the specific packing material that is being replaced.

Installation should be performed by a qualified technician. The following paragraphs will provide. For further details, refer to Packing Installation, Operation and Maintenance Manual.

#### 5.1.3.1 Preparation

##### 5.1.3.1.1 Stem

Inspect stem for any nicks or scratches on the surface finish. Reject the stem for any of these reasons as they may damage packing.

**Note:** *A properly etched part number on the stem in the packing area will have no adverse effect on the performance of the packing.*

Stem finish should be 3-7 AARH (Ra 0,1/0,2).

##### 5.1.3.1.2 Packing Box

**Note:** *Bonnets that have a lube hole or leak detection port are unacceptable for use with the packing arrangement.*

## CAUTION

Packing box should be clean and free of burrs, rust, and any foreign matter. Parts can be cleaned with denatured alcohol.

**Note:** *Packing box finish should be 125 AARH (Ra 3,2) or better.*

The packing box may be bored or honed oversize by up to 0.015" (0.38 mm) above the nominal diameter to improve the finish. For instance, a nominal 0.875" (22.22 mm) packing box may be bored or honed up to 0.890" (22.60 mm) and the "Low-E" Packing will still seal properly.

**Packing box must be finished to the bottom of the bore.**

##### 5.1.3.1.3 Packing

Inspect packing rings. DO NOT use packing if any nicks or scratches on packing are observed. Check packing instructions to ensure that it is in proper arrangement (different packing materials contain arrangements specific to the design).

##### 5.1.3.2 Packing Installation

- a. Refer to the packing instructions that were provided with the packing for proper installation.
- b. Packing Installation, Operation and Maintenance Manual is also available as an additional aid while completing the packing installation.

- c. The packing should be checked for leakage.

**Note: All exposed surfaces of the packing set must be covered with the lubricant.**

- d. Packing load should be checked after the valve has cycled approximately 500 times. Adjust if necessary. Plant maintenance and operation personnel should periodically inspect valves to spot leakage. Adjust if necessary per manufacturer recommendations. If the leak is not resolved, packing and any non-confirming hardware part must be replaced.

#### 5.1.4 Packing Box Efficiency

The regular checking of the packing box is one of the main routing operations of the maintenance service. In order to provide a correct functioning of the valve, the packing should not be tightened over the compression value sufficient to provide the tightness. The packing box efficiency is obtained by the compression of the packing, or by combining it with use of the lubricant. As it wears, gradually re-tighten the packing up to the limits of the possible compression. In order to add packing, it is necessary only to remove the packing follower and flange and introduce one or two split rings.

**Note: In case of emergency, the braided packing of suitable section can be inserted without removal of the worn rings. Before this operation, the valve must be put out of service. If the packing is made up of non-split rings, it may be necessary to disassemble the valve, remove the worn packing, and follow directions to reassemble.**

## 5.2 Repair of Parts

Before reassembly, the parts must be carefully examined in order to eliminate any scratched, worn or damaged parts.

### 5.2.1 Guiding Surfaces

The guiding surfaces of the cage and valve plug, the guide bushing, and the guiding surfaces of the plug stem and auxiliary pilot plug must be particularly checked. If there is only slight damage, a light abrasive can be used.

Otherwise the part must be replaced as soon as possible (see paragraph 1.4 "Spare parts" on page 8).

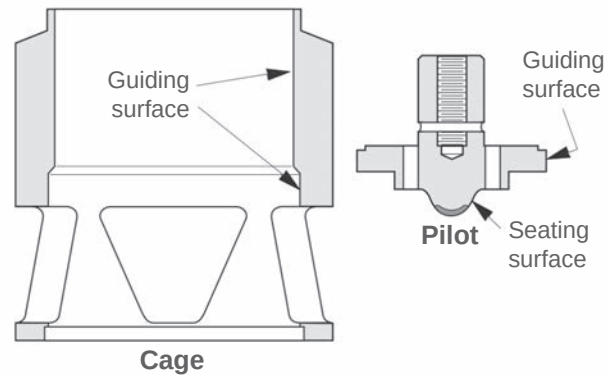


Figure 2 - Cage and Valve Plug

### 5.2.2 Seating Surfaces

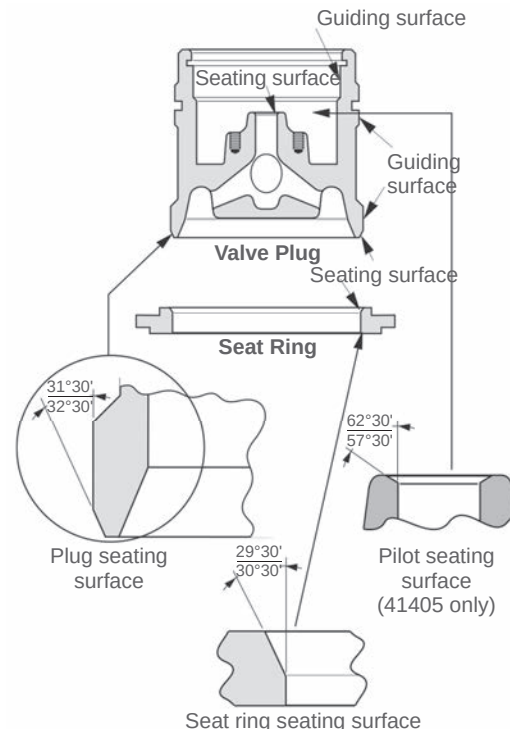


Figure 3 - Seating Surfaces

Inspect all seating surfaces per Figure 3 which represents all plug and seat types.

For type 41405 valves, if the surface of the auxiliary pilot plug (20) is damaged, the auxiliary pilot plug must be replaced (see paragraph 1.4 "Spare parts" on page 8).



The seat ring (13), plug and auxiliary pilot plug (20) seating surfaces must be completely free of dents, wear and scratches.

If the auxiliary pilot plug seat in the valve plug (15) and other valve plug or seat ring seating surfaces show signs of slight deterioration, they can be touched up on a lathe.

For all valve types, when dressing up critical surfaces, no more than 0.25 mm (0.010") of metal must be removed in the case of a 50, 80 or 100 mm (2", 3" or 4") valve and no more than 0.4 mm (0.015") in the case of a 150 to 400 mm (6" to 16") valve. Make sure that the seating angle indicated in Figure 3 on page 14 is respected.

If a slight defect exists on any of the above seating surfaces, lapping can be envisaged. In this case, these instructions must be followed:

- Spread a fine layer of high quality sealing compound on the seating surface.
- Put the seat ring (13) in the body, noting the angle.
- In the case of 41305, 41505, 41605 and 41905 valves, put the cage (16) on the seat ring.
- In case of 41375 valves, put the stack (48) on the seat ring and the cage (16) on the stack.
- In the case of 41405 and 41705 valves, assemble the valve plug, cage and stem (15)-(16)-(1).
- In case of 41305, 41505, 41605, 41375 and 41905, assemble the valve plug (15) and stem (1).
- Assemble the bonnet (7) and its guide bushing (22).
- Place an appropriate tool on the valve plug stem (1) to turn it.
- Lap by slightly rotating the valve plug or the auxiliary pilot plug in alternative directions. After several rotations, lift the valve plug, turn it 90°, and start again.
- Lapping can be repeated, but must be limited as much as possible so that the seat remains sufficiently narrow to guarantee tightness.
- Disassemble the parts, clean them and put them back, respecting the initial angle.

### 5.2.3 Gasket seating surfaces

Gasket seating surface must be free of dents, scratches and corrosion; otherwise, they will need to be repaired.

### 5.2.4 Seal rings and gaskets

Spiral-wound gaskets (10), (14) and (36) must always be changed after disassembly. Seal rings (11A), (11C) or (11E) can be reused if they are free of scratches, erosion and corrosion.

### 5.2.5 Valve plug, valve plug stem

If the valve plug has to be changed, then the stem must also be completely changed to guarantee correct pinning of the assembly. If only the valve plug stem has to be changed, then the valve plug can be reused.

## 6. Valve Reassembly

### 6.1 Pinning the Valve Plug Stem

The valve plug (15) and stem (1) assembly consists of a stem threaded into the valve plug and pinned in place.

If the valve plug (15) [or the auxiliary pilot plug (20) in the case of a 41405 valve] needs to be replaced, it is recommended to replace the stem. The hole of the original pin in an old stem gives bad results and can seriously impair the mechanical strength of the stem-valve assembly.

Assembly is carried out as follows:

#### 6.1.1 Reference Marking on the Valve Plug Stem

Make a reference mark on the valve plug stem at a distance "X" (Table 1 and Figure 5 on page 16), equivalent to the stem recess in the valve plug.

**Note:** *For shouldered plug stems with two flats, the above marking is unnecessary.*

#### 6.1.2 Tightening of the plug stem

To carry out this operation, the valve plug must be prevented from moving by holding the plug shank with an appropriate tool.

Screw two nuts on to the end of the new plug stem and lock them together. Screw the valve plug stem solidly into the plug, checking that the reference mark is level with the end of the plug shank.

If the stem has flats, apply a torque "T" using a wrench of dimension "E" (see Table 1 on page 16).

#### 6.1.3 Drilling the pin hole

(Table 1 and Figure 5 on page 16)

**Note:** *For this operation, it is recommended clamping the valve plug-stem assembly to the plug shank to avoid damaging the guiding surfaces. Particular care must be taken so that the pin hole goes through the valve plug axis.*

If the valve plug is new, drill a hole of diameter "C" at a distance "D" from the end of the valve plug; choose the diameter "C" from the table according to the type of pin used (metric or English pin). If the hole is already drilled in the valve plug, use the hole as a guide to drill through the valve plug stem.

#### 6.1.4 Pinning

By means of a hammer, introduce the pin into the hole. Complete the pinning operation, taking care to ensure that the pin is recessed by the same amount at both sides.

Using a ball tooling and hammer, caulk the pin hole edge of the plug.

Place the assembly in the soft jaw chuck of a lathe to check alignment of the two parts; correct any alignment defects.

## 6.2 Assembly of Ring or Spring-Energized Seal Ring

### 6.2.1 41305 Valve

(Figure 4 on page 16, Figure 9 on page 24, and Figure 12 on page 25)

These valves have spring-energized seal rings composed of a PTFE-based jacket maintained expanding by a spring.

To insert the ring into the plug groove:

- Lubricate the entry chamber.
- Place the ring over the conical top of the plug so that the lips of the ring face the pressure side - see Figure 4.

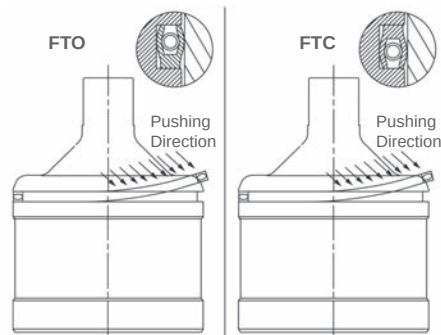


Figure 4 - Spring Energized Seal Ring

## CAUTION

**To the assembly direction of the radial ring (11F) on the plug (see Figure 4). Flow Tends to Open: the open side is in the top. Flow Tends to Close: the open side is in the lower part.**

- Slip the ring into the groove (without spinning it) as shown in Figure 4. This operation can be facilitated by heating the ring. The temperature of the ring should not exceed 150°C.
- Keep the ring retained until it returns to room temperature and goes back in place within the groove. Clamping with a collar (SERFLEX type) will help to properly position the ring.

### 6.2.2 41405 and 41505 and 41705 Valves

(Figures 11 and 12 on pages page 24 and page 25)

These valves are equipped with metal rings; the inner ring has a straight cut while the outer ring has a staggered cut.

To insert the rings into the cage groove, open the rings slightly by hand and slide them, one after the other, along the plug making sure that the parts are not damaged.

**Note: The cuts in the outer and inner rings are to be placed 180° from each other.**

### 6.2.3 41605 Valve (Figure 12 on page 25)

These valves are equipped with an inner elastomeric ring and a PTFE gasket.

Insert the elastomeric ring (11D) in the groove.

Place the PTFE gasket (11C) at a temperature of 100°C (boiling water) to 160°C for a few minutes to facilitate insertion, then slide along the plug until it slips into the groove.

Table 1 - Valve Plug Reference Marking

| Valve plug stem diameter B | Diameter A  | Metric pin, diameter C | Anglo-Saxon pin dia. C | F         | D           | Distance X  | E         | Torque T            |                 |
|----------------------------|-------------|------------------------|------------------------|-----------|-------------|-------------|-----------|---------------------|-----------------|
|                            |             |                        |                        |           |             |             |           | Non-Shouldered stem | Shouldered stem |
| mm (in.)                   | mm (in.)    | mm                     | In.                    | mm (in.)  | mm (in.)    | mm (in.)    | mm (in.)  | N.m (Ft.lbs)        | N.m (Ft.lbs)    |
| 12.7 (.5)                  | 20 (.79)    | 3.5                    | 0.14                   | 18 (.70)  | 18.5 (.73)  | 6 (.24)     | 17 (.67)  | 50 (37)             | 60 (44)         |
| 15.87 (.62)                | 25.5 (1)    | 5                      | 0.2                    | 24 (.95)  | 28 (1.1)    | 8 (.30)     | 22 (.87)  | 50 (37)             | 160 (118)       |
| 19.05 (.75)                | 35 (1.38)   | 5                      | 0.2                    | 30 (1.2)  | 45 (1.77)   | 19 (.75)    | 27 (1.06) | 160 (118)           | 160 (118)       |
| 25.4 (.99)                 | 44.5 (1.66) | 8                      | 0.31                   | 40 (1.58) | 47.5 (1.88) | 25 (.98)    | 30 (1.18) | 160 (118)           | 250 (184)       |
| 31.75 (1.25)               | 58 (2.28)   | 8                      | 0.31                   | 55 (2.20) | 70 (2.76)   | 31.5 (1.24) | 40 (1.57) | -                   | 800 (590)       |
| 38.1 (1.5)                 | 70 (2.76)   | 10                     | -                      | 65 (2.56) | 90 (3.54)   | 33 (1.30)   | 50 (1.97) | -                   | 1 500 (1100)    |

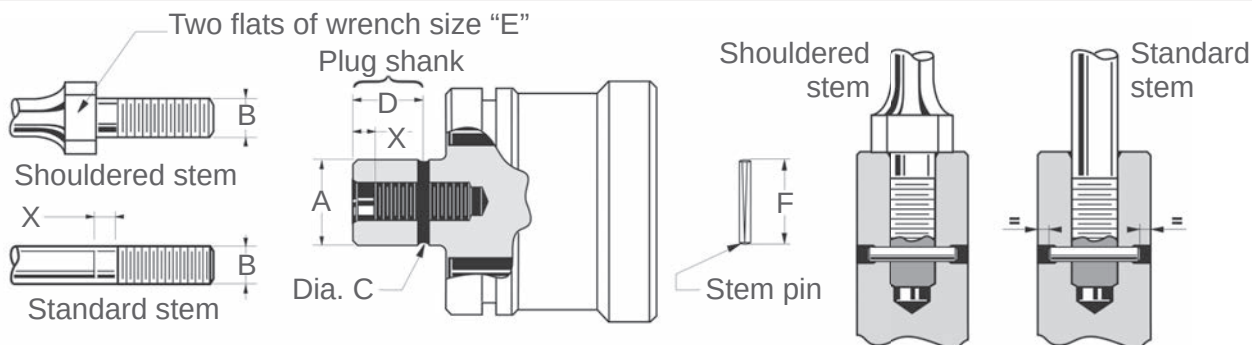


Figure 5 - Plug Stem Marking



For optimum insertion of the ring, a Serflex type ring compressor can be used to compress the ring in the groove for several minutes.

#### 6.2.4 41905 Valve (Figure 12 on page 25)

These valves are equipped with rings; the inner metal ring has a straight cut and the outer ring is made of graphite.

The new graphite seal ring (11A) is supplied in the form of a closed ring in which a notch must be cut before being inserted.

## CAUTION

**Graphite seal rings are fragile and the following operations must be carried out very carefully.**

Using a sharp blade, cut two V notches on the two opposite faces. Hold the ring on either side of the notch between the thumb and index and bend until it breaks.

Using a very fine file, adjust each end of the ring so that its external circumference corresponds to the internal circumference of the inside diameter of the cage (16).

To adjust the length of the ring correctly, insert the new graphite ring into the cage, with the ring against the inner wall of the hole in the cage, allowing minimum play between the two ends of the ring.

To insert the inner ring, then the graphite ring into the cage groove, slightly open the rings by hand and slide them one after the other along the plug, taking care not to damage the parts.

**Note: The cuts on each ring are to be placed 180° from each other.**

#### 6.2.5 41375 High Temperature Valve (232°C to 316°C) (Figure 12 on page 25)

These valves are equipped with a seal ring composed of a jacket and a metallic spring and two backup rings (11G).

**Assemble the different parts as shown in Figure 11 on page 24, starting by the backup rings.**

## CAUTION

**Put these parts as shown in Figure 12 on page 25, the angle of 90° of the backup ring in front of the extrusion gap between cage and plug.**

### 6.3 41405 Valve Plug and Cage Assembly (Figure 10 on page 24)

#### 6.3.1 Assembly of Valve Plug and Auxiliary Pilot Plug

Assemble, as appropriate, the flat spring or the coil springs then the valve plug/plug stem assembly.

Using the same tools as those used for disassembly

(see chapter "Disassembly"), compress the springs so that the retaining ring can be inserted in the groove of the main plug.

#### 6.3.2 Assembly of cage

Place the cage over the valve plug assembly via the top of the plug stem. When doing so, take particular care to position the ring correctly.

### 6.4 Assembly of the Parts Inside the Valve Body

(Figures 7, 8, 9, 10, 11 on page 24)

Proceed as follows:

- After checking that the surfaces are clean, place the seat gasket (14) in the valve body, making sure that it is centered as well as possible.
- Mount the seat ring or the diffuser seat (13).
- Mount:
  - The cage (16) for 41305, 41605 and 41905 valves,
  - The cage/plug/stem assembly for 41405 valves,
  - The stack assembly (making sure that the face with the smallest number of holes is in contact with the seat ring), and cage (16) for 41375 valves.

## CAUTION

For the double-cage assembly, carry out the following operations:

- Put the inner cage (16) upside down,
- Encase the outer cage (75) on the inner cage, maintain them together with the two pins (76).
- In the case of valves other than 41405 valves, insert the valve plug/stem/ring assembly into the cage taking particular care as it goes past the ring or spring-energized seal ring.
- In the case of valves with a nominal diameter less than 150 (6"), place the body/cage gasket in the valve body, making sure that it is centered as well as possible.
- In the case of valve sizes 80 and 100 mm (3" and 4") for high temperature application, insert the flat spring washer (17), then place the body gasket (10) into the valve body, making sure that it is centered as well as possible.
- For valve sizes 150 to 400 mm (6 to 16"), insert either the cage gasket (24) or the flat spring washer (17).
- For valve sizes 450 to 750 mm (18 to 30"), insert either the first cage gasket, the cage washer and the second cage gasket or the flat spring depending on the type of valve, then place the body gasket (10) in the valve body making sure that it is centered as well as possible.
- For valve sizes 800 and 900 mm (32" and 36"), insert the flat spring washer, then place the body gasket (10) in the valve body, making sure that it is centered as well as possible.

# CAUTION

Orientation of the conical spring shall be as follows:

- Nominal diameter 150 (6") to 450 (18") valves, the concave face is upwards
- Nominal diameter 500 (20") to 900 (36") valves, the concave face is downwards and the tapped holes are visible

## 6.5 Assembly of Bonnet

Check that the packing ring (6), the spacer (5) and the guide bushing (22) have been removed from the bonnet.

Position the bonnet (7) above the valve, so that the packing flange studs (2) are perpendicular to the flow direction of the valve.

Lower the bonnet (7) onto the valve stem (1) and push it down carefully until it goes into the valve body studs (21) and is in correct position.

Grease the threads of the valve body studs (21) and the bearing surfaces of the body stud nuts (8).

Screw on the body stud nuts by hand. Tighten the nuts lightly and evenly so that the internal parts are held in place. The face of the bonnet should be parallel to the upper face of the body.

Slide the guide bushing (22) onto the valve plug stem and let it drop to the bottom of the packing box housing.

## 6.6 Tightening of Body Stud Nuts

### Alignment of internal parts

In order to achieve perfect alignment of the seat ring and the valve plug, a force must be applied to the plug stem during tightening of the bonnet which results in correct relative positioning of the two parts.

The force can be applied with the pneumatic actuator as follows:

Place the actuator on the valve bonnet (7) by means of the yoke nut (33) or attachment screws and connect the valve plug stem to the actuator stem. See chapter on actuators for installation instructions.

### Align the internal parts as follows:

In the case of air-to-extend actuators, supply air to the actuator at the maximum pressure indicated on the serial plate and in the case of spring-to-extend actuators, do not supply air to the actuator, so that the optimum positioning of the valve plug and seat can be obtained.

### Single bolting circle:

Evenly tighten the nuts (8) by applying the torque indicated in Tables 2 through 9 on pages page 20 through page 23, and the tightening sequence in Figure 6 on page 19.

### Double bolting circle:

Beginning with the internal nuts, manually screw all the bolts. Screw the internal nuts in order indicated in Figure 6 and screw them by successive, uniform and progressive levels. During the tightening, make sure that the bonnet face stays parallel to the body. When the torque values given in the following Table 2 on page 20 are reached, the bonnet face must be in contact with the body. Screw the external nuts in the order indicated in Figure 6 and screw them by successive, uniform and progressive levels until the torque values given in the tables on next pages are reached.

# CAUTION

During this operation, make sure that the plug does not turn when it is seated. If the plug travel is very small and there is a large amount of plug stem inside the actuator, it may be necessary to remove the yoke nut and lift the actuator so that the plug is not touching the seat.

## 6.7 Assembly of Packing Box

To assemble the packing box, proceed as indicated in the Maintenance chapter; paragraph 5.1.1 on page 12 or 5.1.4 on page 14.

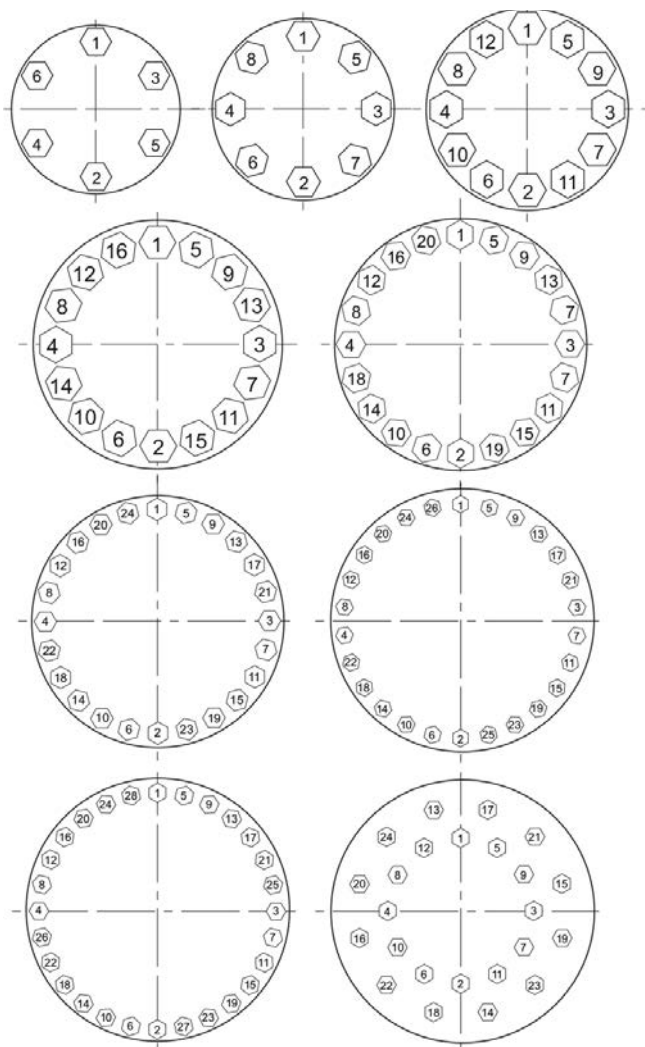


Figure 6 - Tightening Sequence of Nuts (8)

Table 2 - Torque Requirements: Body / Bonnet Bolting [ft-lb]

| Body               |              | Bolts         |          | Torques in ft-lb                                                                                                                                                                                                                   |                                                                                                  |                                                                                                         |
|--------------------|--------------|---------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)    | Quantity | - body/bonnet: all materials<br>- bolts materials :<br>A 193 Gr B7 / A194 Gr 2H,<br>EN 1.7225 / A194 Gr 2H,,<br>A 193 Gr B7M / A194 Gr 2HM,<br>A 320 Gr L7 / A194 Gr 7,<br>A 193 Gr B16 / A194 Gr 7 or<br>A 453 Gr 660 / A194 Gr 7 | - body/bonnet material:<br>stainless steel<br>- bolts materials:<br>A 453 Gr 660 /<br>A 194 GR 8 | - body/bonnet material:<br>stainless steel<br>- bolts materials:<br>A 193 Gr B8 class 2 /<br>A 194 GR 8 |
| 2                  | 300          | 3/4-10UNC-2A  | 6        | 103                                                                                                                                                                                                                                | 162                                                                                              | 162                                                                                                     |
| 3 x 2              | 600          | 3/4-10UNC-2A  | 6        | 162                                                                                                                                                                                                                                | 258                                                                                              | 258                                                                                                     |
| 4 x 2              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 2                  | 900          | 7/8-9UNC-2A   | 8        | 155                                                                                                                                                                                                                                | 243                                                                                              |                                                                                                         |
|                    | 1500         | 7/8-9UNC-2A   | 8        | 221                                                                                                                                                                                                                                | 354                                                                                              |                                                                                                         |
|                    | 2500         | 1-1/4-8-UN-2A | 6        | 406                                                                                                                                                                                                                                | 649                                                                                              |                                                                                                         |
| 3 x 2              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 4 x 2              | 900 - 1500   | 1-8UN-2A      | 8        | 332                                                                                                                                                                                                                                | 531                                                                                              |                                                                                                         |
| 3                  | 300          | 3/4-10UNC-2A  | 8        | 133                                                                                                                                                                                                                                | 214                                                                                              | 214                                                                                                     |
|                    | 600          | 3/4-10UNC-2A  | 8        | 162                                                                                                                                                                                                                                | 258                                                                                              | 258                                                                                                     |
|                    | 900          | 1-1/4-8-UN-2A | 6        | 561                                                                                                                                                                                                                                | 900                                                                                              |                                                                                                         |
|                    | 1500         | 1-1/4-8-UN-2A | 6        | 811                                                                                                                                                                                                                                | 1306                                                                                             |                                                                                                         |
| 4 x 3              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 6 x 3              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 3                  | 2500         | 1-1/2-8UN-2A  | 6        | 774                                                                                                                                                                                                                                | 1239                                                                                             |                                                                                                         |
| 4                  | 300          | 7/8-9UNC-2A   | 8        | 170                                                                                                                                                                                                                                | 273                                                                                              |                                                                                                         |
|                    | 600          | 7/8-9UNC-2A   | 8        | 258                                                                                                                                                                                                                                | 413                                                                                              |                                                                                                         |
|                    | 900          | 1-1/2-8UN-2A  | 6        | 885                                                                                                                                                                                                                                | 1431                                                                                             |                                                                                                         |
|                    | 1500         | 1-1/2-8UN-2A  | 6        | 1365                                                                                                                                                                                                                               | 2205                                                                                             |                                                                                                         |
| 6 x 4              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 8 x 4              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 4                  | 2500         | 1-5/8-8UN-2A  | 8        | 922                                                                                                                                                                                                                                | 1497                                                                                             |                                                                                                         |
| 6                  | 300          | 1-8UN-2A      | 8        | 325                                                                                                                                                                                                                                | 516                                                                                              |                                                                                                         |
|                    | 600          | 1-8UN-2A      | 12       | 236                                                                                                                                                                                                                                | 376                                                                                              |                                                                                                         |
|                    | 900 - 1500   | 1-3/4-8UN-2A  | 8        | 1365                                                                                                                                                                                                                               | 2220                                                                                             |                                                                                                         |
| 8 x 6              |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 10 x 6             |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 6                  | 2500         | 1-7/8-8UN-2A  | 8        | 1401                                                                                                                                                                                                                               | 2279                                                                                             |                                                                                                         |
| 8                  | 300          | 1-1/4-8-UN-2A | 8        | 524                                                                                                                                                                                                                                | 841                                                                                              |                                                                                                         |
|                    | 600          | 1-1/4-8-UN-2A | 12       | 479                                                                                                                                                                                                                                | 774                                                                                              |                                                                                                         |
|                    | 900          | 1-3/4-8UN-2A  | 8        | 1365                                                                                                                                                                                                                               | 2220                                                                                             |                                                                                                         |
|                    | 1500         | 1-3/4-8UN-2A  | 8        | 1770                                                                                                                                                                                                                               | 2877                                                                                             |                                                                                                         |
| 10 x 8             |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 12 x 8             |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 8                  | 2500         | 1-3/4-8UN-2A  | 12       | 1217                                                                                                                                                                                                                               | 1992                                                                                             |                                                                                                         |
| 10                 | 300          | 1-1/2-8UN-2A  | 8        | 848                                                                                                                                                                                                                                | 1372                                                                                             |                                                                                                         |
|                    | 600          | 1-1/2-8UN-2A  | 12       |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
|                    | 900          | 1-3/4-8UN-2A  | 12       | 1365                                                                                                                                                                                                                               | 2220                                                                                             |                                                                                                         |
|                    | 1500         | 1-3/4-8UN-2A  | 12       | 1623                                                                                                                                                                                                                               | 2633                                                                                             |                                                                                                         |
|                    | 2500         | 2-8UN-2A      | 12       | 2065                                                                                                                                                                                                                               | 3363                                                                                             |                                                                                                         |
| 12                 | 300          | 1-1/2-8UN-2A  | 8        | 922                                                                                                                                                                                                                                | 1490                                                                                             |                                                                                                         |
|                    | 600          | 1-1/2-8UN-2A  | 12       | 848                                                                                                                                                                                                                                | 1372                                                                                             |                                                                                                         |
|                    | 900          | 1-1/2-8UN-2A  | 16       | 922                                                                                                                                                                                                                                | 1490                                                                                             |                                                                                                         |
|                    | 1500         | 1-3/4-8UN-2A  | 16       | 1623                                                                                                                                                                                                                               | 2633                                                                                             |                                                                                                         |
| 16 x 12            |              |               |          |                                                                                                                                                                                                                                    |                                                                                                  |                                                                                                         |
| 12                 | 2500         | 1-7/8-8UN-2A  | 20       | 1623                                                                                                                                                                                                                               | 2641                                                                                             |                                                                                                         |
| 14                 | 300          | 1-1/2-8UN-2A  | 8        | 1564                                                                                                                                                                                                                               | 1778                                                                                             |                                                                                                         |
|                    | 600          | 1-1/2-8UN-2A  | 12       | 1564                                                                                                                                                                                                                               | 1778                                                                                             |                                                                                                         |
|                    | 900          | 2-8UN-2A      | 16       | 3275                                                                                                                                                                                                                               | 3717                                                                                             |                                                                                                         |
|                    | 1500         | 2-1/2-8UN-2A  | 16       | 6609                                                                                                                                                                                                                               | 7501                                                                                             |                                                                                                         |
| 16                 | 300          | 1-1/2-8UN-2A  | 12       | 848                                                                                                                                                                                                                                | 1372                                                                                             |                                                                                                         |
|                    | 600          | 1-1/2-8UN-2A  | 16       | 922                                                                                                                                                                                                                                | 1490                                                                                             |                                                                                                         |
|                    | 900          | 1-1/2-8UN-2A  | 20       | 1106                                                                                                                                                                                                                               | 1792                                                                                             |                                                                                                         |
|                    | 1500         | 1-3/4-8UN-2A  | 20       | 1918                                                                                                                                                                                                                               | 3113                                                                                             |                                                                                                         |
|                    | 2500         | 2-8UN-2A      | 24       | 1955                                                                                                                                                                                                                               | 3186                                                                                             |                                                                                                         |
| 18                 | 300          | 1-3/8-8UN-2A  | 16       | 679                                                                                                                                                                                                                                | 1092                                                                                             |                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 16       | 1512                                                                                                                                                                                                                               | 2456                                                                                             |                                                                                                         |
|                    | 900          | 2-1/4-8UN-2A  | 12       | 3356                                                                                                                                                                                                                               | 5480                                                                                             |                                                                                                         |
|                    | 1500         | 2-1/2-8UN-2A  | 16       | 4794                                                                                                                                                                                                                               | 7848                                                                                             |                                                                                                         |
| 20 <sup>(1)</sup>  | 300          | 1-3/8-8UN-2A  | 24       | 738                                                                                                                                                                                                                                | 1202                                                                                             |                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 24       | 1549                                                                                                                                                                                                                               | 2515                                                                                             |                                                                                                         |
|                    | 900          | 1-7/8-8UN-2A  | 24       | 2950                                                                                                                                                                                                                               | 4802                                                                                             |                                                                                                         |
| 24                 | 300          | 1-3/4-8UN-2A  | 20       | 1401                                                                                                                                                                                                                               | 2279                                                                                             |                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 26       | 1770                                                                                                                                                                                                                               | 2877                                                                                             |                                                                                                         |
|                    | 900          | 2-1/2-8UN-2A  | 20       | 5163                                                                                                                                                                                                                               | 8453                                                                                             |                                                                                                         |

1. 20" 41005-GS requires 20 bolts for the 300# and 600#. Size of bolts and torque requirements do not change.

**Table 3 - Torque Requirements: Body / Bonnet Bolting [ft-lb] - sizes 32 and 36in**

| Body               |              | Bolts        |          | Torques in ft-lb                                                                                                                                                                                                                                  |                                                                                                                          |                                                                                                                          |
|--------------------|--------------|--------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)   | Quantity | - <u>body/bonnet</u> : all materials<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H,<br>EN 1.7225 / A194 Gr 2H,,<br>A 193 Gr B7M / A194 Gr 2HM,<br>A 320 Gr L7 / A194 Gr 7,<br>A 193 Gr B16 / A194 Gr 7 or<br>A 453 Gr 660 / A194 Gr 7 | - <u>body/bonnet material</u> :<br><u>stainless steel</u><br>- <u>bolts materials</u> :<br>A 453 Gr 660 /<br>A 194 GR 8. | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B8 class 2 /<br>A 194 GR 8. |
| 32                 | 150          | 1-7/8-8UN-2A | 24       | 1400                                                                                                                                                                                                                                              | 2213                                                                                                                     |                                                                                                                          |
|                    | 300          | 1-7/8-8UN-2A | 32       | 1770                                                                                                                                                                                                                                              | 2876                                                                                                                     |                                                                                                                          |
| 36                 | 150          | 1-7/8-8UN-2A | 32       | 1460                                                                                                                                                                                                                                              | 2374                                                                                                                     |                                                                                                                          |
|                    | 300          | 1-7/8-8UN-2A | 36       | 1961                                                                                                                                                                                                                                              | 3090                                                                                                                     |                                                                                                                          |

**Table 4 - Torque Requirements: Body / Bonnet Bolting [ft-lb] 410A5, 410B5, or 410C5 High Capacity Design**

| Body               |              | Bolts         |          | Torques in ft-lb                                                                                                                                                                                                                                  |                                                                                                                                              |                                                                                                                         |
|--------------------|--------------|---------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)    | Quantity | - <u>body/bonnet</u> : all materials<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H,<br>EN 1.7225 / A194 Gr 2H,,<br>A 193 Gr B7M / A194 Gr 2HM,<br>A 320 Gr L7 / A194 Gr 7,<br>A 193 Gr B16 / A194 Gr 7 or<br>A 453 Gr 660 / A194 Gr 7 | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H<br>A 453 Gr 660 / A 194<br>GR 8 | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B8 class 2 /<br>A 194 GR 8 |
| 6                  | 150/300      | 1-8UN-2A      | 10       | 332                                                                                                                                                                                                                                               | 376                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-8UN-2A      | 12       | 317                                                                                                                                                                                                                                               | 354                                                                                                                                          |                                                                                                                         |
| 8                  | 150/300      | 1-1/4-8-UN-2A | 10       | 465                                                                                                                                                                                                                                               | 546                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-1/4-8-UN-2A | 12       | 487                                                                                                                                                                                                                                               | 546                                                                                                                                          |                                                                                                                         |
| 10                 | 150/300      | 1-1/2-8UN-2A  | 8        | 1254                                                                                                                                                                                                                                              | 1269                                                                                                                                         |                                                                                                                         |
|                    | 600          | 1-1/2-8UN-2A  | 10       | 1313                                                                                                                                                                                                                                              | 1342                                                                                                                                         |                                                                                                                         |
| 12                 | 150/300      | 1-3/4-8UN-2A  | 12       | 810                                                                                                                                                                                                                                               | 906                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 12       | 1214                                                                                                                                                                                                                                              | 1360                                                                                                                                         |                                                                                                                         |
| 16                 | 150/300      | 1-3/4-8UN-2A  | 10       | 1984                                                                                                                                                                                                                                              | 2021                                                                                                                                         |                                                                                                                         |
|                    | 600          | 2-8UN-2A      | 10       | 3135                                                                                                                                                                                                                                              | 3208                                                                                                                                         |                                                                                                                         |
| 18                 | 150/300      | 1-1/2-8UN-2A  | 20       | 826                                                                                                                                                                                                                                               | 937                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 20       | 1785                                                                                                                                                                                                                                              | 2021                                                                                                                                         |                                                                                                                         |
| 20                 | 150/300      | 1-3/4-8UN-2A  | 20       | 1401                                                                                                                                                                                                                                              | 2279                                                                                                                                         |                                                                                                                         |
| 24                 | 150/300      | 2-1/2-8UN-2A  | 12       | 5236                                                                                                                                                                                                                                              | 5089                                                                                                                                         |                                                                                                                         |

**Table 5 - Torque Requirements: Body / Bonnet Bolting [ft-lb] – HD Design - sizes 3 and 4in**

| Body               |              | Bolts        |          | Torques in ft-lb                                                                                                                    |                                                                                                                                     |                                                                                                    |                                                                                                       |
|--------------------|--------------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)   | Quantity | - <u>body/bonnet</u> :<br>A217 Gr WC6<br>1.7357 EN 10213<br>A217 Gr WC9<br>- <u>bolts materials</u> :<br>A193 Gr B16 /<br>A194 Gr 7 | - <u>body/bonnet</u> :<br>A217 Gr WC6<br>1.7357 EN 10213<br>A217 Gr WC9<br>- <u>bolts materials</u> :<br>B637 N07718 /<br>A194 Gr 8 | - <u>body/bonnet</u> :<br>A217 Gr C12A<br>- <u>bolts materials</u> :<br>B637 N07718 /<br>A194 Gr 8 | - <u>body/bonnet</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A453 Gr 660 /<br>A194 Gr 8 |
| 3                  | 300/600      | 7/8"-9UNC-2A | 10       | 177                                                                                                                                 | 280                                                                                                                                 | 612                                                                                                | 273                                                                                                   |
|                    | 900/1500     | 1"3/8-8UN-2A | 8        | 760                                                                                                                                 | 1232                                                                                                                                | 2434                                                                                               | 1195                                                                                                  |
| 4                  | 300/600      | 7/8"-9UNC-2A | 12       | 184                                                                                                                                 | 295                                                                                                                                 | 605                                                                                                | 288                                                                                                   |
|                    | 900/1500     | 1"3/8-8UN-2A | 10       | 841                                                                                                                                 | 1364                                                                                                                                | 2434                                                                                               | 1320                                                                                                  |

**Table 6 - Torque Requirements: Body / Bonnet Bolting [m.daN]**

| Body                  |              | Bolts         |        | Torques in m.daN                                                                                                                                                                                                                                  |                                                                                                                  |                                                                                                                         |
|-----------------------|--------------|---------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Nominal sizes (in)    | ASME Classes | Sizes (in)    | Number | - <u>body/bonnet</u> : all materials<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H,<br>EN 1.7225 / A194 Gr 2H,,<br>A 193 Gr B7M / A194 Gr 2HM,<br>A 320 Gr L7 / A194 Gr 7,<br>A 193 Gr B16 / A194 Gr 7 or<br>A 453 Gr 660 / A194 Gr 7 | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 453 Gr 660 /<br>A 194 GR 8 | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B8 class 2 /<br>A 194 GR 8 |
| 2                     | 300          | 3/4-10UNC-2A  | 6      | 14                                                                                                                                                                                                                                                | 22                                                                                                               | 22                                                                                                                      |
| 3 x 2<br>4 x 2        | 600          | 3/4-10UNC-2A  | 6      | 22                                                                                                                                                                                                                                                | 35                                                                                                               | 35                                                                                                                      |
| 2                     | 900          | 7/8-9UNC-2A   | 8      | 21                                                                                                                                                                                                                                                | 33                                                                                                               |                                                                                                                         |
|                       | 1500         | 7/8-9UNC-2A   | 8      | 30                                                                                                                                                                                                                                                | 48                                                                                                               |                                                                                                                         |
|                       | 2500         | 1-1/4-8-UN-2A | 6      | 55                                                                                                                                                                                                                                                | 88                                                                                                               |                                                                                                                         |
| 3 x 2<br>4 x 2        | 900 - 1500   | 1-8UN-2A      | 8      | 45                                                                                                                                                                                                                                                | 72                                                                                                               |                                                                                                                         |
| 3<br>4 x 3<br>6 x 3   | 300          | 3/4-10UNC-2A  | 8      | 18                                                                                                                                                                                                                                                | 29                                                                                                               | 29                                                                                                                      |
|                       | 600          | 3/4-10UNC-2A  | 8      | 22                                                                                                                                                                                                                                                | 35                                                                                                               | 35                                                                                                                      |
|                       | 900          | 1-1/4-8-UN-2A | 6      | 76                                                                                                                                                                                                                                                | 122                                                                                                              |                                                                                                                         |
|                       | 1500         | 1-1/4-8-UN-2A | 6      | 110                                                                                                                                                                                                                                               | 177                                                                                                              |                                                                                                                         |
| 3                     | 2500         | 1-1/2-8UN-2A  | 6      | 105                                                                                                                                                                                                                                               | 168                                                                                                              |                                                                                                                         |
| 4<br>6 x 4<br>8 x 4   | 300          | 7/8-9UNC-2A   | 8      | 23                                                                                                                                                                                                                                                | 37                                                                                                               |                                                                                                                         |
|                       | 600          | 7/8-9UNC-2A   | 8      | 35                                                                                                                                                                                                                                                | 56                                                                                                               |                                                                                                                         |
|                       | 900          | 1-1/2-8UN-2A  | 6      | 120                                                                                                                                                                                                                                               | 194                                                                                                              |                                                                                                                         |
|                       | 1500         | 1-1/2-8UN-2A  | 6      | 185                                                                                                                                                                                                                                               | 299                                                                                                              |                                                                                                                         |
| 4                     | 2500         | 1-5/8-8UN-2A  | 8      | 125                                                                                                                                                                                                                                               | 203                                                                                                              |                                                                                                                         |
| 6<br>8 x 6<br>10 x 6  | 300          | 1-8UN-2A      | 8      | 44                                                                                                                                                                                                                                                | 70                                                                                                               |                                                                                                                         |
|                       | 600          | 1-8UN-2A      | 12     | 32                                                                                                                                                                                                                                                | 51                                                                                                               |                                                                                                                         |
|                       | 900 - 1500   | 1-3/4-8UN-2A  | 8      | 185                                                                                                                                                                                                                                               | 301                                                                                                              |                                                                                                                         |
| 6                     | 2500         | 1-7/8-8UN-2A  | 8      | 190                                                                                                                                                                                                                                               | 309                                                                                                              |                                                                                                                         |
| 8<br>10 x 8<br>12 x 8 | 300          | 1-1/4-8-UN-2A | 8      | 71                                                                                                                                                                                                                                                | 114                                                                                                              |                                                                                                                         |
|                       | 600          | 1-1/4-8-UN-2A | 12     | 65                                                                                                                                                                                                                                                | 105                                                                                                              |                                                                                                                         |
|                       | 900          | 1-3/4-8UN-2A  | 8      | 185                                                                                                                                                                                                                                               | 301                                                                                                              |                                                                                                                         |
|                       | 1500         | 1-3/4-8UN-2A  | 8      | 240                                                                                                                                                                                                                                               | 390                                                                                                              |                                                                                                                         |
| 8                     | 2500         | 1-3/4-8UN-2A  | 12     | 165                                                                                                                                                                                                                                               | 270                                                                                                              |                                                                                                                         |
| 10                    | 300          | 1-1/2-8UN-2A  | 8      | 115                                                                                                                                                                                                                                               | 186                                                                                                              |                                                                                                                         |
|                       | 600          | 1-1/2-8UN-2A  | 12     |                                                                                                                                                                                                                                                   |                                                                                                                  |                                                                                                                         |
|                       | 900          | 1-3/4-8UN-2A  | 12     | 185                                                                                                                                                                                                                                               | 301                                                                                                              |                                                                                                                         |
|                       | 1500         | 1-3/4-8UN-2A  | 12     | 220                                                                                                                                                                                                                                               | 357                                                                                                              |                                                                                                                         |
|                       | 2500         | 2-8UN-2A      | 12     | 280                                                                                                                                                                                                                                               | 456                                                                                                              |                                                                                                                         |
| 12<br>16 x 12         | 300          | 1-1/2-8UN-2A  | 8      | 125                                                                                                                                                                                                                                               | 202                                                                                                              |                                                                                                                         |
|                       | 600          | 1-1/2-8UN-2A  | 12     | 115                                                                                                                                                                                                                                               | 186                                                                                                              |                                                                                                                         |
|                       | 900          | 1-1/2-8UN-2A  | 16     | 125                                                                                                                                                                                                                                               | 202                                                                                                              |                                                                                                                         |
|                       | 1500         | 1-3/4-8UN-2A  | 16     | 220                                                                                                                                                                                                                                               | 357                                                                                                              |                                                                                                                         |
| 12                    | 2500         | 1-7/8-8UN-2A  | 20     | 220                                                                                                                                                                                                                                               | 358                                                                                                              |                                                                                                                         |
| 14                    | 300          | 1-1/2-8UN-2A  | 8      | 212                                                                                                                                                                                                                                               | 241                                                                                                              |                                                                                                                         |
|                       | 600          | 1-1/2-8UN-2A  | 12     | 212                                                                                                                                                                                                                                               | 241                                                                                                              |                                                                                                                         |
|                       | 900          | 2-8UN-2A      | 16     | 444                                                                                                                                                                                                                                               | 504                                                                                                              |                                                                                                                         |
|                       | 1500         | 2-1/2-8UN-2A  | 16     | 896                                                                                                                                                                                                                                               | 1017                                                                                                             |                                                                                                                         |
| 16                    | 300          | 1-1/2-8UN-2A  | 12     | 115                                                                                                                                                                                                                                               | 186                                                                                                              |                                                                                                                         |
|                       | 600          | 1-1/2-8UN-2A  | 16     | 125                                                                                                                                                                                                                                               | 202                                                                                                              |                                                                                                                         |
|                       | 900          | 1-1/2-8UN-2A  | 20     | 150                                                                                                                                                                                                                                               | 243                                                                                                              |                                                                                                                         |
|                       | 1500         | 1-3/4-8UN-2A  | 20     | 260                                                                                                                                                                                                                                               | 422                                                                                                              |                                                                                                                         |
|                       | 2500         | 2-8UN-2A      | 24     | 265                                                                                                                                                                                                                                               | 432                                                                                                              |                                                                                                                         |
| 18                    | 300          | 1-3/8-8UN-2A  | 16     | 92                                                                                                                                                                                                                                                | 148                                                                                                              |                                                                                                                         |
|                       | 600          | 1-3/4-8UN-2A  | 16     | 205                                                                                                                                                                                                                                               | 333                                                                                                              |                                                                                                                         |
|                       | 900          | 2-1/4-8UN-2A  | 12     | 455                                                                                                                                                                                                                                               | 743                                                                                                              |                                                                                                                         |
|                       | 1500         | 2-1/2-8UN-2A  | 16     | 650                                                                                                                                                                                                                                               | 1064                                                                                                             |                                                                                                                         |
| 20 <sup>(1)</sup>     | 300          | 1-3/8-8-UN 2A | 24     | 100                                                                                                                                                                                                                                               | 163                                                                                                              |                                                                                                                         |
|                       | 600          | 1-3/4-8UN-2A  | 24     | 210                                                                                                                                                                                                                                               | 341                                                                                                              |                                                                                                                         |
|                       | 900          | 1-7/8-8UN-2A  | 24     | 400                                                                                                                                                                                                                                               | 651                                                                                                              |                                                                                                                         |
| 24                    | 300          | 1-3/4-8UN-2A  | 20     | 190                                                                                                                                                                                                                                               | 309                                                                                                              |                                                                                                                         |
|                       | 600          | 1-3/4-8UN-2A  | 26     | 240                                                                                                                                                                                                                                               | 390                                                                                                              |                                                                                                                         |
|                       | 900          | 2-1/2-8UN-2A  | 20     | 700                                                                                                                                                                                                                                               | 1146                                                                                                             |                                                                                                                         |

1. 20" 41005-GS requires 20 bolts for the 300# and 600#. Size of bolts and torque requirements do not change.



**Table 7 - Torque Requirements: Body / Bonnet Bolting [m.daN] - sizes 32 and 36in**

| Body               |              | Bolts        |          | Torques in m. daN                                                                                                                                                                                                     |                                                                                                                   |                                                                                                                          |
|--------------------|--------------|--------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)   | Quantity | - <u>body/bonnet</u> : all materials<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H,<br>EN 1.7225 / A194 Gr 2H,,<br>A 193 Gr B7M / A194 Gr 2HM,<br>A 320 Gr L7 / A194 Gr 7,<br>A 193 Gr B16 / A194 Gr 7 or | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 453 Gr 660 /<br>A 194 GR 8. | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B8 class 2 /<br>A 194 GR 8. |
| 32                 | 150          | 1-7/8-8UN-2A | 24       | 190                                                                                                                                                                                                                   | 300                                                                                                               | N/A                                                                                                                      |
|                    | 300          | 1-7/8-8UN-2A | 32       | 240                                                                                                                                                                                                                   | 390                                                                                                               | N/A                                                                                                                      |
| 36                 | 150          | 1-7/8-8UN-2A | 32       | 198                                                                                                                                                                                                                   | 322                                                                                                               | N/A                                                                                                                      |
|                    | 300          | 1-7/8-8UN-2A | 36       | 266                                                                                                                                                                                                                   | 419                                                                                                               | N/A                                                                                                                      |

**Table 8 - Torque Requirements: Body / Bonnet Bolting [m.daN] 410A5, 410B5, or 410C5 High Capacity Design**

| Body               |              | Bolts         |          | Torques in m.daN                                                                                                                                                                                                                                  |                                                                                                                                              |                                                                                                                         |
|--------------------|--------------|---------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)    | Quantity | - <u>body/bonnet</u> : all materials<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H,<br>EN 1.7225 / A194 Gr 2H,,<br>A 193 Gr B7M / A194 Gr 2HM,<br>A 320 Gr L7 / A194 Gr 7,<br>A 193 Gr B16 / A194 Gr 7 or<br>A 453 Gr 660 / A194 Gr 7 | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B7 / A194 Gr 2H<br>A 453 Gr 660 / A 194<br>GR 8 | - <u>body/bonnet material</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A 193 Gr B8 class 2 /<br>A 194 GR 8 |
| 6                  | 150/300      | 1-8UN-2A      | 10       | 45                                                                                                                                                                                                                                                | 51                                                                                                                                           |                                                                                                                         |
|                    | 600          | 1-8UN-2A      | 12       | 43                                                                                                                                                                                                                                                | 48                                                                                                                                           |                                                                                                                         |
| 8                  | 150/300      | 1-1/4-8-UN-2A | 10       | 63                                                                                                                                                                                                                                                | 74                                                                                                                                           |                                                                                                                         |
|                    | 600          | 1 1/8 8 UN 2A | 12       | 66                                                                                                                                                                                                                                                | 74                                                                                                                                           |                                                                                                                         |
| 10                 | 150/300      | 1-1/2-8UN-2A  | 8        | 170                                                                                                                                                                                                                                               | 172                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-1/2-8UN-2A  | 10       | 178                                                                                                                                                                                                                                               | 182                                                                                                                                          |                                                                                                                         |
| 12                 | 150/300      | 1-3/4-8UN-2A  | 12       | 110                                                                                                                                                                                                                                               | 123                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 12       | 165                                                                                                                                                                                                                                               | 177                                                                                                                                          |                                                                                                                         |
| 16                 | 150/300      | 1-3/4-8UN-2A  | 10       | 269                                                                                                                                                                                                                                               | 274                                                                                                                                          |                                                                                                                         |
|                    | 600          | 2-8UN-2A      | 10       | 425                                                                                                                                                                                                                                               | 435                                                                                                                                          |                                                                                                                         |
| 18                 | 150/300      | 1-1/2-8UN-2A  | 20       | 112                                                                                                                                                                                                                                               | 127                                                                                                                                          |                                                                                                                         |
|                    | 600          | 1-3/4-8UN-2A  | 20       | 242                                                                                                                                                                                                                                               | 274                                                                                                                                          |                                                                                                                         |
| 20                 | 150/300      | 1-3/4-8UN-2A  | 20       | 190                                                                                                                                                                                                                                               | 309                                                                                                                                          |                                                                                                                         |
| 24                 | 150/300      | 2-1/2-8UN-2A  | 12       | 710                                                                                                                                                                                                                                               | 690                                                                                                                                          |                                                                                                                         |

**Table 9 - Torque Requirements: Body / Bonnet Bolting [m.daN] - HD Design- sizes 3 and 4in**

| Body               |              | Bolts        |          | Torques in m. daN                                                                                                                   |                                                                                                                                     |                                                                                                    |                                                                                                       |
|--------------------|--------------|--------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Nominal sizes (in) | ASME Classes | Sizes (in)   | Quantity | - <u>body/bonnet</u> :<br>A217 Gr WC6<br>1.7357 EN 10213<br>A217 Gr WC9<br>- <u>bolts materials</u> :<br>A193 Gr B16 /<br>A194 Gr 7 | - <u>body/bonnet</u> :<br>A217 Gr WC6<br>1.7357 EN 10213<br>A217 Gr WC9<br>- <u>bolts materials</u> :<br>B637 N07718 /<br>A194 Gr 8 | - <u>body/bonnet</u> :<br>A217 Gr C12A<br>- <u>bolts materials</u> :<br>B637 N07718 /<br>A194 Gr 8 | - <u>body/bonnet</u> :<br>stainless steel<br>- <u>bolts materials</u> :<br>A453 Gr 660 /<br>A194 Gr 8 |
| 3                  | 300/600      | 7/8-9UNC-2A  | 10       | 24                                                                                                                                  | 38                                                                                                                                  | 83                                                                                                 | 37                                                                                                    |
|                    | 900/1500     | 1-3/8-8UN-2A | 8        | 103                                                                                                                                 | 167                                                                                                                                 | 330                                                                                                | 162                                                                                                   |
| 4                  | 300/600      | 7/8-9UNC-2A  | 12       | 25                                                                                                                                  | 40                                                                                                                                  | 82                                                                                                 | 39                                                                                                    |
|                    | 900/1500     | 1-3/8-8UN-2A | 10       | 114                                                                                                                                 | 185                                                                                                                                 | 330                                                                                                | 179                                                                                                   |



## 7. Body Sub-Assembly

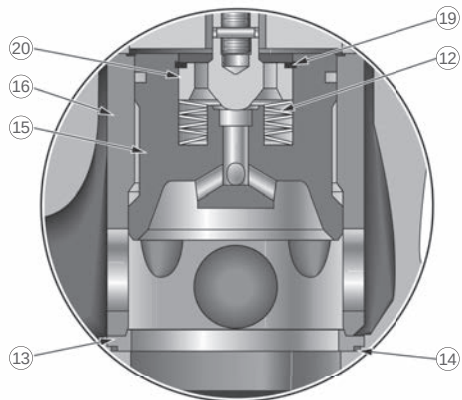


Figure 7 - 41405 Trim: 2" to 4"

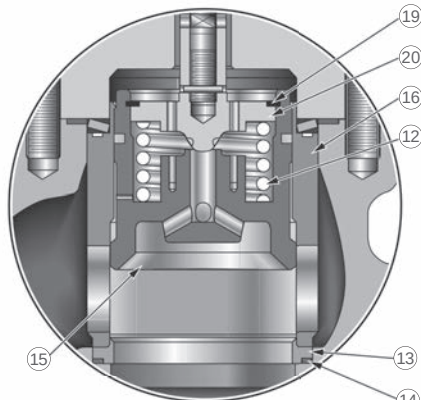


Figure 8 - 41405 Trim: 6" to 18"

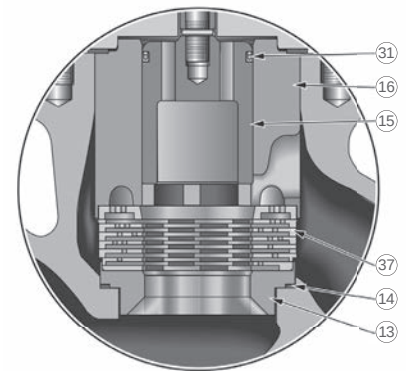
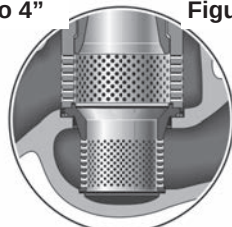


Figure 9 - Internal parts of VRT anticavitation valves – 41375 type

Figure 10 - Internal parts of pilot plug valve – 41405 type



Diffuser Option

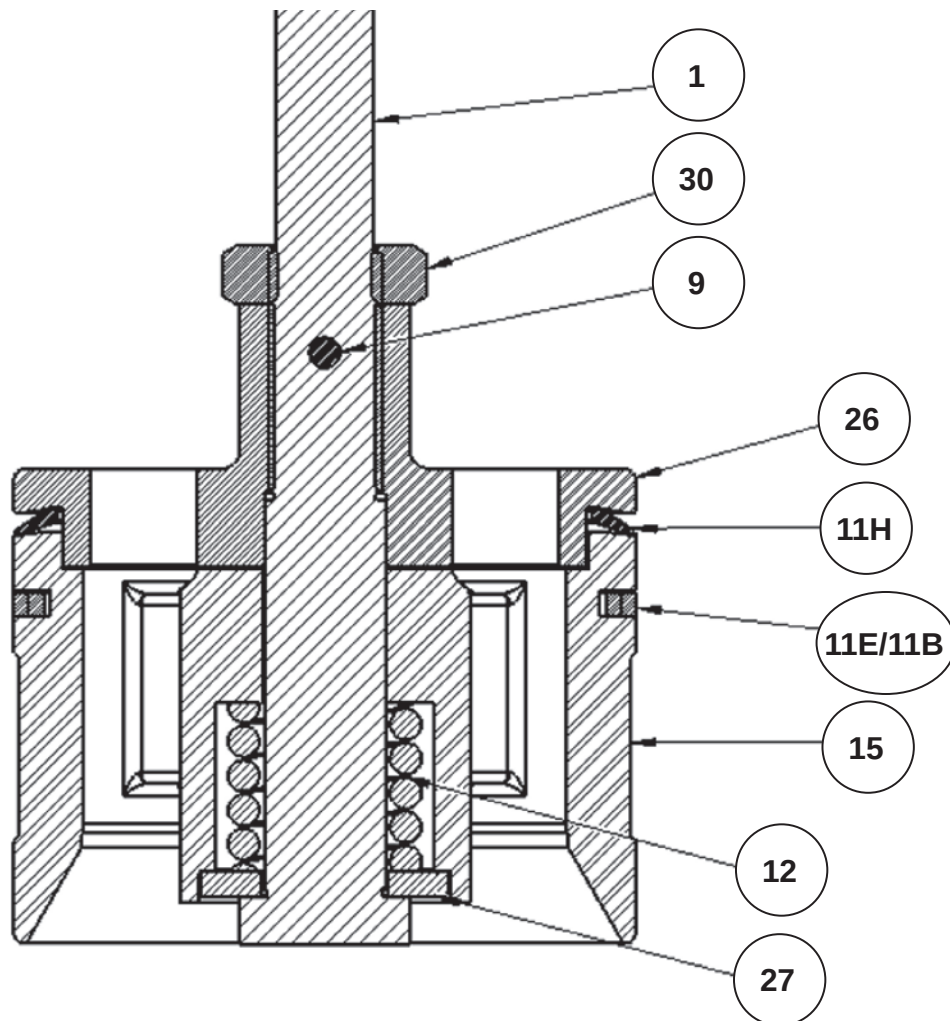
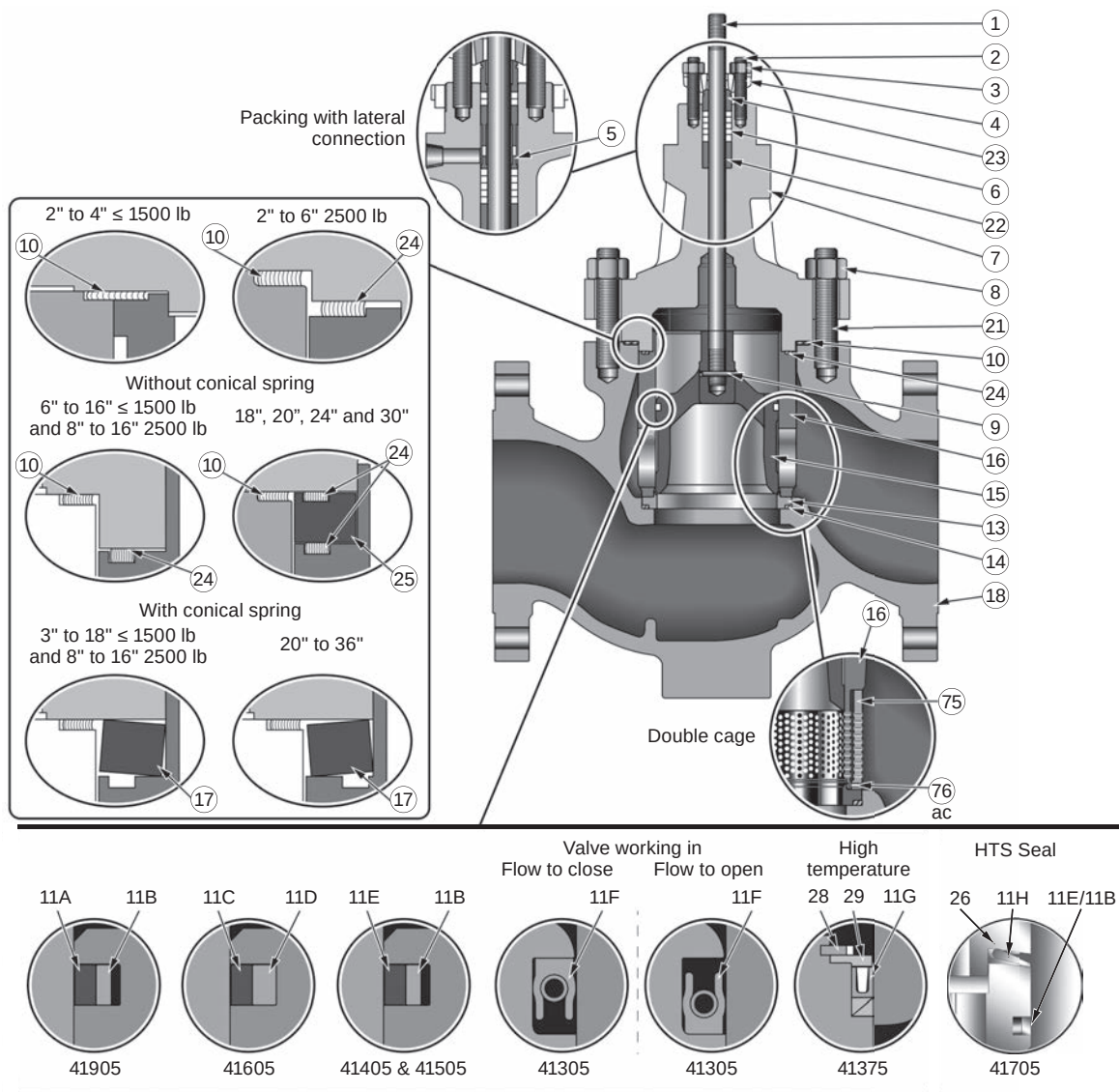


Figure 11 - HTS plug assembly



**Figure 12 - General Assembly and Bill of Materials**

| Parts Reference                                  |   |     |                     |  |                                   |     |             |                                |      |                                         |             |             |                    |
|--------------------------------------------------|---|-----|---------------------|--|-----------------------------------|-----|-------------|--------------------------------|------|-----------------------------------------|-------------|-------------|--------------------|
| Ref.                                             |   |     | Designation         |  | Ref.                              |     | Designation |                                | Ref. |                                         | Designation |             |                    |
|                                                  |   | 1   | Valve plug stem     |  | ▲ ●                               | 11F | Tec seal    |                                |      |                                         | 24          | Cage Gasket |                    |
|                                                  |   | 2   | Packing flange stud |  | ○                                 |     | 11G         | Seal ring and backup ring      |      |                                         |             | 25          | Cage washer        |
|                                                  |   | 3   | Packing flange nut  |  |                                   |     | 11H         | HTS seal                       |      |                                         |             | 26          | Upper plug         |
|                                                  |   | 4   | Packing flange      |  | ❖                                 |     | 12          | Spring (or spring washers set) |      |                                         |             | 27          | Spring washer      |
|                                                  |   | 5   | Packing spacer      |  |                                   |     | 13          | Seat/diffuser seat             |      | ○                                       |             | 28          | Retaining ring     |
|                                                  | ● | 6   | Packing             |  |                                   | ●   | 14          | Seat ring gasket               |      | ○                                       |             | 29          | Retaining ring     |
|                                                  |   | 7   | Bonnet              |  |                                   |     | 15          | Valve plug                     |      |                                         |             | 30          | Jam nut            |
|                                                  |   | 8   | Valve body nut      |  |                                   |     | 16          | Cage                           |      |                                         |             | 37          | Stack              |
|                                                  | ● | 9   | Plug stem pin       |  | ◆                                 |     | 17          | Spring washer                  |      |                                         |             | 50          | Washer (body nuts) |
|                                                  | ● | 10  | Body gasket         |  | ❖                                 |     | 18          | Valve body                     |      |                                         |             | 75          | Double cage        |
| □                                                | ● | 11A | Graphite seal ring  |  |                                   |     | 19          | Retaining ring                 |      |                                         |             | 76          | Pin                |
| □                                                | ● | 11B | Metallic ring       |  | ❖                                 |     | 20          | Auxiliary pilot plug           |      |                                         |             |             |                    |
| ★                                                | ● | 11C | U PTFE seal ring    |  |                                   |     | 21          | Valve body stud                |      |                                         |             |             |                    |
| ★                                                | ● | 11D | Elastomer seal ring |  |                                   |     | 22          | Guide bushing                  |      |                                         |             |             |                    |
| ■                                                | ● | 11E | Metallic seal ring  |  |                                   |     | 23          | Packing follower               |      |                                         |             |             |                    |
| ❖ On 41405 series valves only                    |   |     |                     |  | ● Recommended spare part          |     |             |                                |      | ▲ On 41305 series valves only           |             |             |                    |
| ◆ On 6" to 24" valves sizes only (150 to 600 mm) |   |     |                     |  | □ On 41905 series valves only     |     |             |                                |      | ○ On 41375 high temperature valves only |             |             |                    |
| ★ On 41605 series valves only                    |   |     |                     |  | ■ On 41405/505 Series valves only |     |             |                                |      |                                         |             |             |                    |

## 8. Actuation

Refer to the instruction, operation and maintenance manuals for the 51/52/53 Series Pneumatic Piston Actuators and the 87/88 Series Actuators.

# Appendix A

## 41005 API-6A Series High Pressure Valves

### A1. General

#### A1.1 Scope

The following instructions are designed to guide the user through the installation and maintenance of the 41005 API 6A Series valves for all sizes and for high working pressures (i.e. 10K PSI and 15K PSI).

For 41005 API 6A Valve Series, Baker Hughes has developed special options which is the subject of this section in the instruction manual. In this case, the instructions given in this section will always take precedence over the general instructions found in the preceding sections.

#### A1.2 Serial Plate

The serial plate is usually fixed to the side of the actuator yoke. It indicates, amongst other things, the type of valve, the working pressure, the material class, temperature class, Product Specification Level, Performance Requirement used for the pressurized chamber and the air supply pressure of the actuator.

There are two serial plates, one for the valve body sub assembly and one for the actuator sub assembly.

#### A1.3 After Sales Service

Baker Hughes offers its customers an After Sales Service comprising highly qualified technicians for the operation, maintenance and repair of its equipment. To benefit from this service, contact your local representative or the After Sales Service whose email address is given at the end of this document.

#### A1.4 Spare Sparts

When carrying out maintenance operations, only manufacturer's replacement parts must be used, obtained through our local representatives or our Spare Parts Service.

When ordering spare parts, the model and serial numbers indicated on the manufacturer's serial plate must be given.

The recommended spare parts are indicated in the parts list included in this Appendix in Table 16 on page 40.

#### A1.5 Actuator and Other Accessories

The valve is equipped with an actuator. Like other valve accessories, actuators have specific instructions regarding the installation, operation and maintenance for the electric and pneumatic connections.

For additional information, please refer to the instruction, operation and maintenance manuals for the 51/52/53 Series Pneumatic Piston Actuators and the 87/88 Series Actuators.

#### A1.6 Warranty

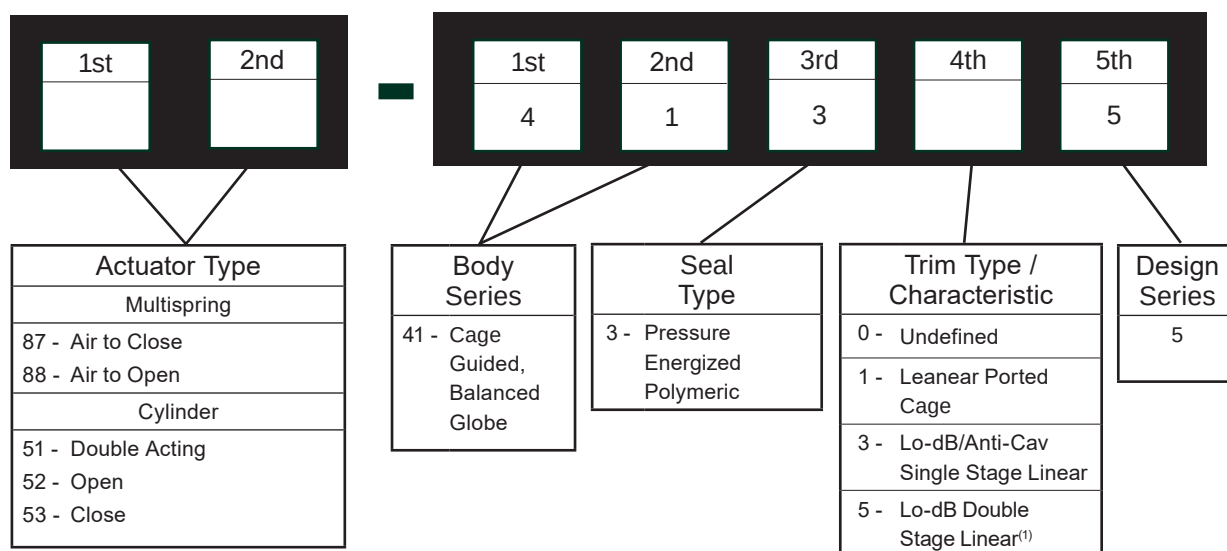
Refer to General terms beginning on page 6 of the Manual.

**41005 API 6A should be closed for a short period only. There is a risk of damage for trim parts for long period of closing.**

**41005 API 6A valves are designed as control valves, not as isolating valves.**

**If 41005 are used as isolating valve, internal parts damage will not be covered by warranty.**

### 41005 API 6A Numbering System



(1) Lo-dB Double stage option is not available for 10" API 6A valves.

## A2. Installation

### A2.1 Cleanness of Piping

Before installing the valve in the line, clean piping and valve of all foreign material such as welding chips, scale, oil, grease or dirt. Gasket surfaces must be thoroughly cleaned to insure leak-free joints.

### A2.2 Isolating By-Pass Valve

To allow for in-line inspection, maintenance and removal of the valve without service interruption, provide a manually operated stop valve on each side of the control valve and a manually operated throttling valve in the by-pass line.

### A2.3 Heat Insulation

In case of heat insulated installation, do not insulate the valve bonnet and take protection measures related to personal safety.

### A2.4 Hydraulic Test and Cleaning of Lines

During this operation, the control valve must not be used as an isolating valve.

This means that the valve must always be opened before carrying out pressure tests in the process line, cleaning of pipes, etc., otherwise equipment damages or destroying of seal rings could result.

### A2.5 Flow Direction

The valve must be installed so that the controlled substance will flow through the valve in the direction indicated by the flow arrow located on the body.

## A3. Disassembly

### A3.1 Removal of Actuator

(Figures 7 and 8 on page 24)

Access to the internal components of the body should be accomplished with the actuator removed. To carry out this operation, follow the instructions below and refer to the specific actuator instructions in the Installation, Operation and Maintenance manuals for actuator types 87/88 and 51/52/53.

#### A3.1.1 Disconnection of instruments

Disconnect all mechanical couplings between the positioner and the other instruments on one side and the valve stem/ actuator stem coupling on the other side.

#### A3.1.2 Plug stems screwed into the actuator stem

In the case of air-to-retract actuators, apply sufficient air pressure on the diaphragm to retract the stem completely. Loosen the counter-nut, unscrew the stem.

## CAUTION

During this operation, make sure that the plug does not turn when it is seated. If the plug travel is very small and there is a large amount of plug stem inside the actuator, it may be necessary to remove the yoke nut and lift the actuator so that the plug is not touching the seat.

#### A3.1.3 Stems attached with a stem connector

In the case of air-to-retract actuators, apply sufficient air pressure on the diaphragm to retract the stem completely.

Loosen the screws and remove the stem connector.

#### A3.1.4 Removal of actuator

Disconnect all the ingoing and outgoing air and electrical connections from the actuator. Loosen the yoke nut or attachment screws and lift the actuator, making sure that the concentricity and the thread of the bonnet is not damaged.

### A3.2 Opening of the Pressurized Chamber (Figures 9, 10 and 11 on page 24)

## ! DANGER

Prior to disassembly, vent the process pressure and isolate the valve if necessary.

**Note:** The valve must always be reassembled with new packing rings and gaskets. Before disassembly, make sure that the appropriate parts are available.

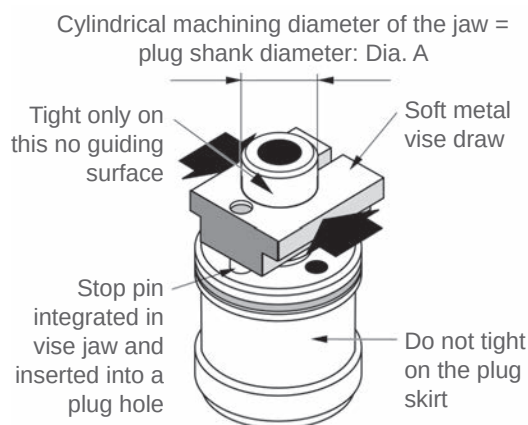


- A. Remove the packing flange nuts (B221) then remove the packing flange/follower (B219).
- B. Check that the exposed part of the valve plug stem (B150) is clean enough for the bonnet (B003) to be removed easily.
- C. Remove the body stud nuts (B014).
- D. By means of a Lifting Lug secured instead of the actuator, lift the bonnet (B003) up and separate it from the valve body (B001). During this operation, the valve plug stem (B150) must be pushed downwards so that the valve plug remains in the valve body (B001).
- E. Remove the the body gasket (B015).
- F. In the case of type 41305 valve, remove the valve plug stem (B150) and valve plug (B112) assembly from the cage by pulling the valve plug stem upward, then remove the cage (B107).
- G. Remove the trim retainer (B105) followed by the cage (B107) and the seat ring gasket (B103) from the valve body (B001).
- H. Remove the packing (B207, B208), and the packing bushing (B203) from the bonnet (B003).

### A3.3 Disassembly of Valve Plug Stem

The valve plug stem is screwed and pinned into the valve plug (B112).

To dismount the stem, the valve plug must be held as indicated below, taking care not to damage the guiding surfaces; the plug stem pin (B903 ) is then removed. By means of flats or using a nut and counter-nut on the end of the stem, unscrew the stem taking care not to apply a bending moment which could deform it.



**Figure 13 - Valve Plug Stem**

## A4. Maintenance and Repair

### A4.1 Preliminary Controls

Before reassembly, check parts and the materials for conformity to the bill of material call out. Inspect inside of body (B001), Bonnet (B003), Cages (B107, B110) and plug (B112) for cleanliness.

Remove any foreign matter and clean the gasket surfaces. Inspect body (B001), Bonnet (B003), Cages (B107, B110), gaskets (B103, B108, B118, B015), plug (B112), plug stems (B150), and bushings (B203) for nicks, scratches, burrs, sharp corners, etc., on sealing and sliding surfaces including packing box.

#### Important:

**During assembly, care must be exercised to prevent damage to these parts.**

Since 41005 API 6A are high pressure valves, ensure there is no damage to the sealing surfaces. Performance of gaskets and seals are strongly linked to surface roughness.

### A4.2 Packing Box

Prior to assembly the packing:

- Ensure there are no nicks or scratches on the stem and packing box inside diameter packing area.

Violation of any of these requirements is cause for stem or packing box rejection.

Tightness of the packing box is obtained by compression of the packing rings (B207) and anti-extrusion rings (B208). Compression must be achieved by evenly tightening the packing flange nuts (B221) on the packing flange (B219). Periodical retightening of the packing flange nuts is required to maintain tightness.

Make sure that the packing is not overtightened as this could prevent smooth operation of the valve.

If a leak persists after the packing has been compressed to a maximum, the packing needs to be changed.

#### A4.2.1 PTFE Based packing

- A. Unscrew and remove the packing flange nut (B221) and packing flat washer (B921a)
- B. Lift the packing flange (B219) up along the valve stem
- C. Extract and remove the packing rings (B207) and anti-extrusion rings (B208), taking care not to damage the sealing surface of the packing box lantern or the valve plug stem.
- D. Replace the packing rings, with the cut in each ring placed about 120° from the next, pressing them down one at a time following instructions in Table 10 on page 30.

**Table 10 - PTFE Based Packing**

| N.D. mm (in.) | Stem Size (in.) | Number of Packing Rings | Number of anti-extrusion rings |
|---------------|-----------------|-------------------------|--------------------------------|
| 80 (3")       | 25.4 (1")       | 4                       | 2                              |
| 100 (4")      | 25.4 (1")       | 4                       | 2                              |
| 150 (6")      | 28.56 (1"1/8)   | 5                       | 2                              |
| 200 (8")      | 31.75 (1"1/4)   | 4                       | 2                              |
| 250 (10")     | 31.75 (1"1/4)   | 4                       | 2                              |

- E. Install Packing studs (B220).
- F. Install the guide bushing (B203) in the bottom of the packing box.
- G. Install packing set (B207 and B208).
- H. Assemble Packing flange/follower (B219).
- I. Screw the packing flange nuts (B221) by hand.
- J. Check that the valve plug (B112) moves freely.
- K. Gradually apply torque values indicated in Table 11.  
Pay attention to keep an even gap between the stem (B150) and packing flange/follower (B219) when torquing. There must be no contact between the stem (B150) and the packing flange/ follower (B219).

**Table 11 - Torque Values**

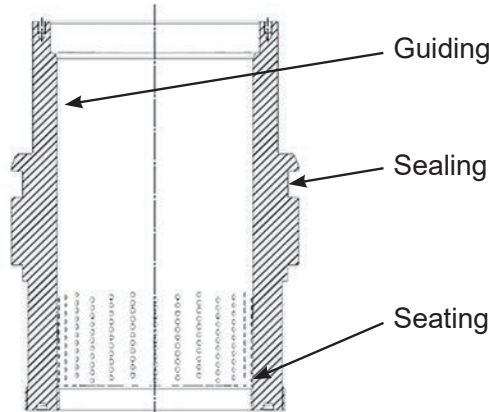
| Description   | Stem Size (inch) | 1             | 1"1/8         | 1"1/4         |
|---------------|------------------|---------------|---------------|---------------|
|               | Valve Class      | 10K/15K       | 10K/15K       | 10K/15K       |
|               | Bolt Thread size | 1/2"-13UNC-2A | 1/2"-13UNC-2A | 5/8"-11UNC-2A |
|               | Bolt number      | 4             | 4             | 4             |
| Packing studs | Torque [N.m]     | 82            | 69            | 151           |
|               | Torque [ft-lb]   | 61            | 51            | 111           |

### A4.3 Repair of Parts

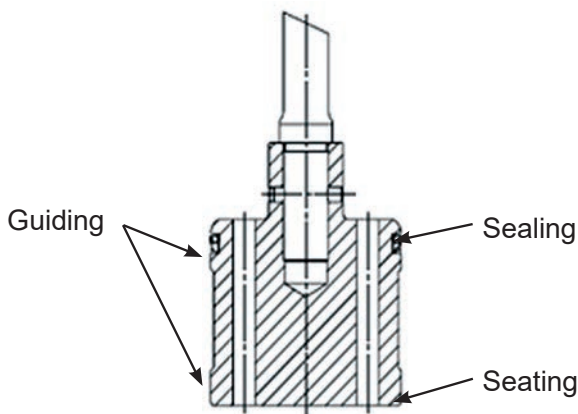
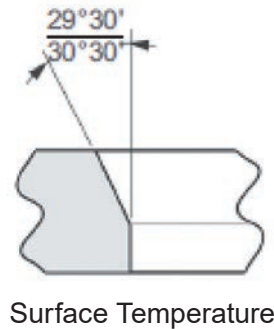
Before re-assembly, the parts must be carefully examined in order to eliminate any scratched, worn or damaged parts.

#### A4.3.1 Guiding Surfaces

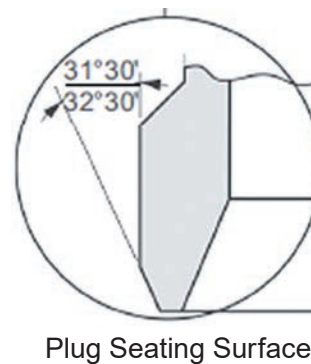
The guiding surfaces of the cage and valve plug, the guide bushing, and the guiding surfaces of the plug stem and packing box must be checked. Otherwise, the part must be replaced as soon as possible (see paragraph 1.4 "Spare Parts" on page 8).



**Figure 14 - Cage Guiding, Sealing & Seating Surfaces**



**Figure 15 - Plug Guiding, Sealing & Seating Surfaces**





### A4.3.2 Seating Surfaces

Inspect all seating surfaces per Figures 14 and 15 on page 30 which represent all plug and seat types.

The cage (B107), plug (B112) seating surfaces must be completely free of dents, wear and scratches.

If valve plug (B112) or cage (B107) seating surfaces show signs of slight deterioration, they can be touched up on a lathe.

For all valve types, when dressing up critical surfaces, no more than 0.25 mm (0.010") of metal must be removed in the case of an 80 or 100 mm (3" or 4") valve, and no more than 0.4 mm (0.015") in the case of a 150 to 250 mm (6" to 10") valve. Make sure that the seating angle indicated in Figures 2 and 3 on page 14 is respected.

If a slight defect exists on any of the above seating surfaces, lapping can be envisaged, in this case, the instructions hereunder must be followed:

1. Spread a fine layer of high-quality sealing compound on the seating surface.
2. Put the cage (B107) in the body, noting the angle.
3. Assemble the valve plug (B112) and Stem (B150).
4. Assemble the bonnet (B003) and its guide bushing (B203).
5. Place an appropriate tool on the valve plug stem (B150) to turn it.
6. Lap by slightly rotating the valve plug (B112) in alternative directions. After several rotations, lift the valve plug (B112), turn it 90°, and start again.
7. Lapping can be repeated but must be limited as much as possible so that the cage (B107) seating area remains sufficiently narrow to guarantee tightness.
8. Disassemble the parts, clean them and put them back in their initial angle.

### A4.3.3 Gasket seating surfaces

Gasket seating surface must be free of dents, scratches and corrosion. If damaged, the gasket seating surface will need to be repaired.

### A4.3.4 Seal Rings and gaskets

Seat ring gaskets (B103), lips seals (B108 and B118), Body gasket (B015) must always be replaced after disassembly.

### A4.3.5 Valve Plug and Plug Stem

If the valve plug (B112) needs to be changed, then the stem must also be completely changed to guarantee correct pinning of the assembly. If only the valve plug stem (B150) is changed, then the valve plug (B112) can be reused.

## A5. Valve Reassembly

### A5.1 Pinning the Valve Plug Stem

The valve plug (B112) and stem (B150) assembly consist of a stem threaded into the valve plug and pinned in place.

If the valve plug (B112) needs to be replaced, a new stem must be installed.

The hole of the original pin in an old stem often prevents satisfactory results being obtained and can seriously impair the mechanical strength of the stem-valve assembly.

#### A5.1.1 Tightening of the plug stem

To carry out this operation, the valve plug must be prevented from moving by holding the plug shank with an appropriate tool.

Screw two nuts on to the end of the new plug stem and lock them together. Screw the valve plug stem solidly into the plug, checking that the reference mark is level with the end of the plug shank.

If the stem has flats, apply a torque "T" using a wrench of dimension "E" (see Table 12 on page 32).

#### A5.1.2 Drilling the Pin Hole

(Figure 15 on page 30 and Table 13 on page 35)

**Note: For this operation, it is recommended clamping the valve plug-stem assembly by the plug shank to avoid damaging the guiding surfaces; particular care must be taken so that the pin hole goes through the valve plug axis.**

If the valve plug (B112) is new, drill a hole of diameter "C" at a distance "D" from the end of the valve plug (B112); choose the diameter "C" from the table according to the type of pin used (metric or English pin). If the hole is already drilled in the valve plug (B112), use the hole as a guide to drill through the valve plug stem (B150).

#### A5.1.3 Pinning

By means of a hammer, insert the pin (B903) into the hole. Complete the pinning operation by ensuring that the pin (B903) is recessed by the same amount on both sides.

Using a ball tooling and hitting with a hammer, proceed with caulking the pin hole edge of the plug (B112). This will prevent the pin from going out.

Place the assembly in the soft jaw chuck of a lathe to check alignment of the two parts; correct any alignment defects.

Table 12 - Pinning and Stem Torque Values

| Pinning and Stem Torque Values |               |                              |                             |              |              |               |           |                  |
|--------------------------------|---------------|------------------------------|-----------------------------|--------------|--------------|---------------|-----------|------------------|
| Valve Plug Stem Diameter<br>B  | Diameter<br>A | Metric pin,<br>Diameter<br>C | Anglo-Saxon<br>pin Diameter | F            | D            | Distance<br>X | E         | Torque T         |
| mm (inch)                      | mm (inch)     | mm                           | inch                        | mm<br>(inch) | mm<br>(inch) | mm<br>(inch)  | mm (inch) | N.m (Ft.<br>lbs) |
| 25.4 (0.99)                    | 42 (1.65)     | 8                            | .31                         | 40 (1.57)    | 28 (1.1)     | 16 (.63)      | 30 (1.18) | 250 (184)        |
| 28.56 (1.125)                  | 42 (1.65)     | 8                            | .31                         | 40 (1.57)    | 47.5 (1.87)  | 16 (.63)      | 36 (1.41) | 250 (184)        |
| 31.75 (1.25)                   | 58 (2.28)     | 8                            | .31                         | 40 (1.57)    | 70 (2.76)    | 32 (1.26)     | 40 (1.57) | 800 (590)        |

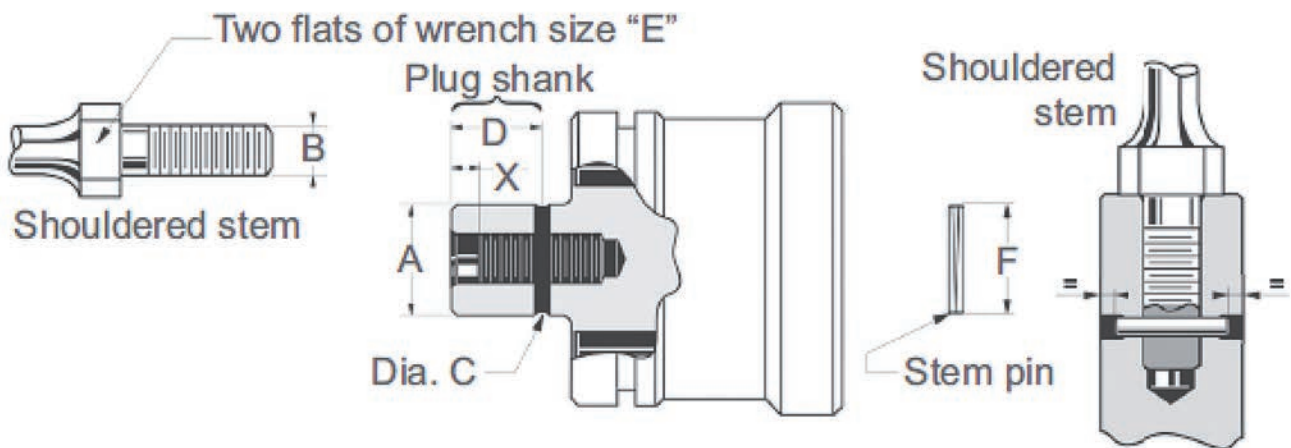


Figure 16 - Valve Plug and Stem Pinning

## A5.2 Assembly of Spring-energized Seal Ring

These valves have spring-energized seal rings maintained expanding by a spring.

Installation of the high-pressure lip seal is like that of the standard lip seal with the exception that it requires additional force due to the added stiffness of the seal. In addition to requiring more force, care must be taken not to damage the lip of the seal during the installation. See Figures 17 on page 32, 18, 19 and 20 on page 33.

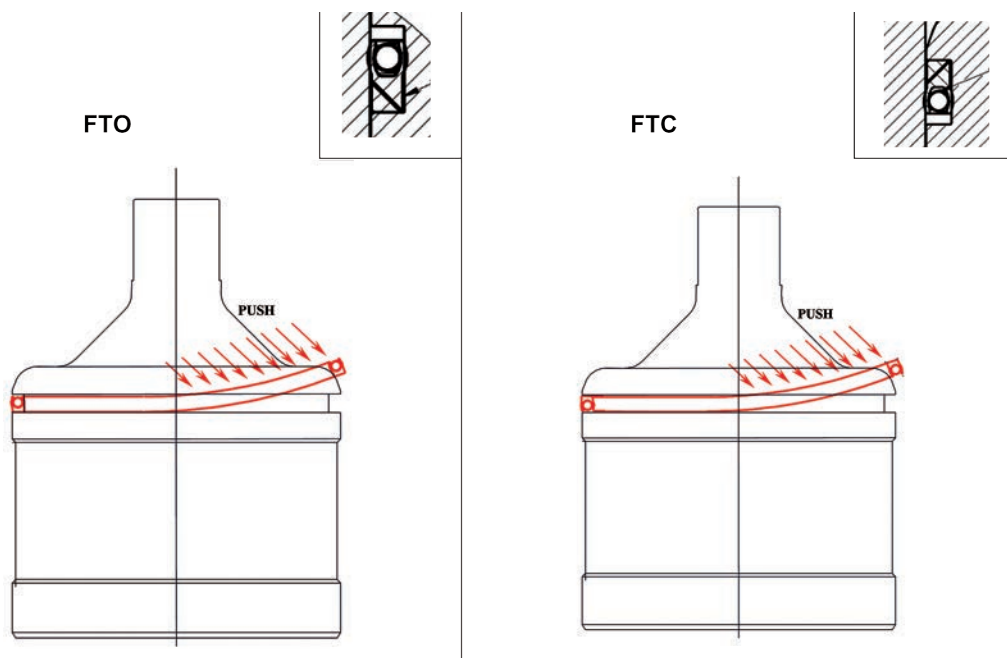
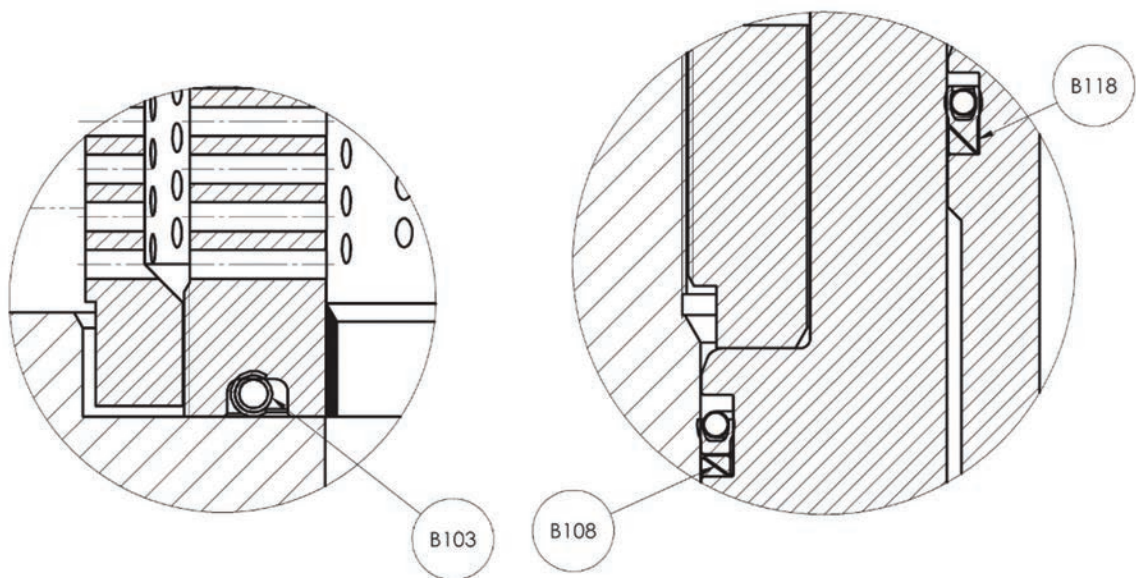
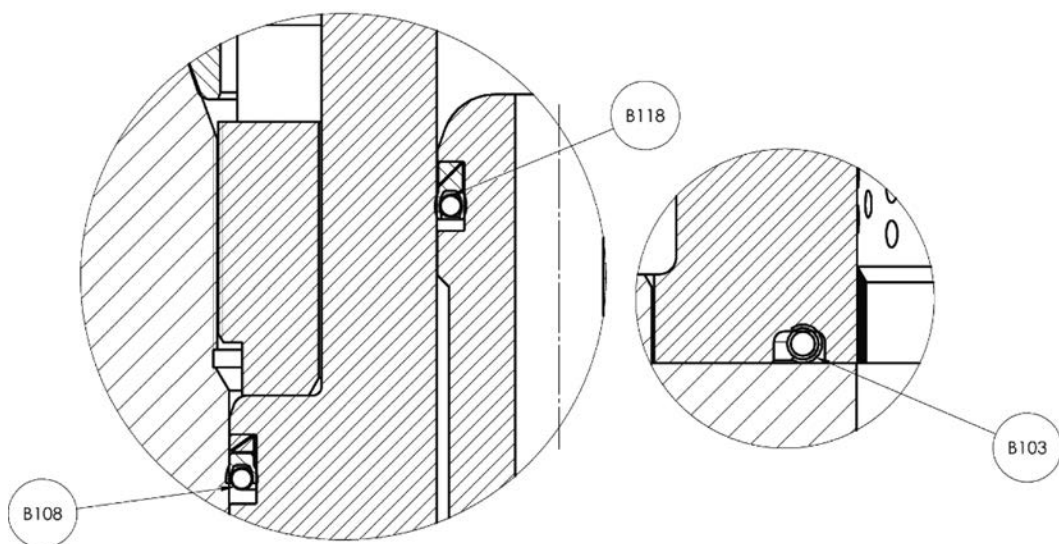


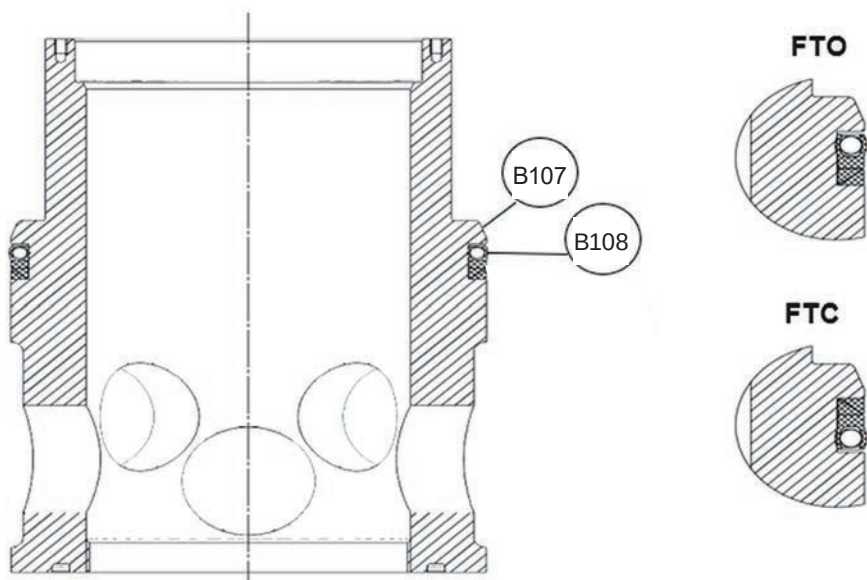
Figure 17 - Plug Balanced Seal Mounting



**Figure 18 - Flow To Open Seals and Gasket Mounting**



**Figure 19 - Flow To Close Seals and Gasket Mounting**



**Figure 20 - Principle Diagram of the Cage Lip Seal Mounting**

To insert rings into groove:

1. Lubricate the entry chamber.
2. Place the ring over the conical top of the plug so that the lips of the ring face the pressure side (see Figures 19 and 20 on page 33).

## CAUTION

**Assemble the direction of radial rings (B108 and B118) on the plug and on the cage (see Figures 18 and 19 on page 33). Flow to open: the open side is in the top. Flow to close: the open side is in the bottom.**

3. Slip the ring into the groove (without spinning it) as shown in Figure 20 on page 33. This operation can be facilitated by heating the ring. The temperature of the ring should not exceed 180°C (356°F).
4. Keep the ring retained until it returns to room temperature and goes back in place within the groove. Clamping with a collar (SERFLEX type) will help to properly position the ring.

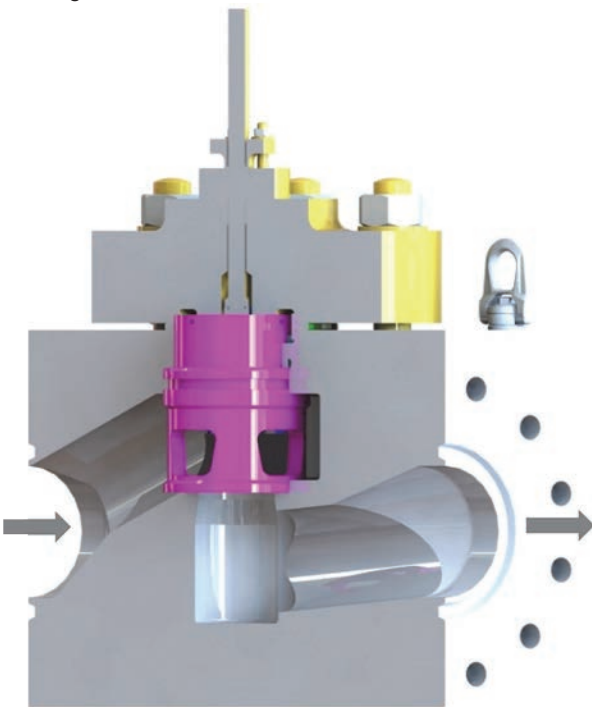


Figure 21 - Flow To Close Direction

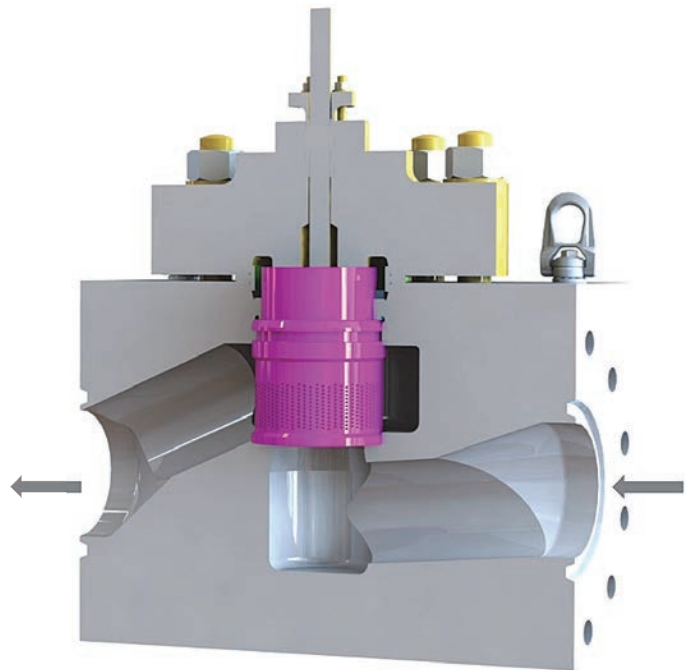


Figure 22 - Flow To Open Direction



### A5.3 Assembly of the Parts Inside the Valve Body

1. Preliminary Controls: refer back to section A4.1 on page 29.
2. Mount the cage (B107).
3. In case of double cage TRIM 41355 API 6A see Table 13, assembly as Figure 23.
  - a. Put the inner cage (13) upside down.
  - b. Screw the outer cage (B110) and the inner cage (B107) at the torque value as per Table 13.
  - c. Screw Sets screws (B914) to the outer cage (B110).

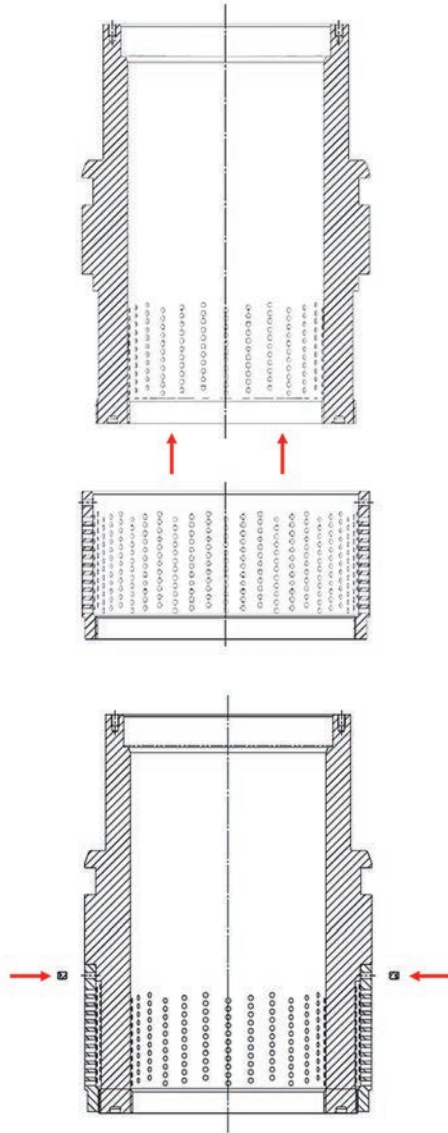


Figure 23 - Exploded View of 41355 API 6A Trim

Table 13 - Torque Value for 41355 API 6A Trim

| Description | Valve Size (inch) | 3             | 4             | 6             | 8             | 10 <sup>(1)</sup> |
|-------------|-------------------|---------------|---------------|---------------|---------------|-------------------|
|             | Valve Class       | 10K/15K       | 10K/15K       | 10K/15K       | 10K/15K       | 10K/15K           |
| Outer/Inner | Thread size       | 4"1/2-16UN-2B | 5"5/8-16UN-2B | 6"3/8-16UN-2B | 8"1/2-16UN-2B | N/A               |
|             | Torque [daN.m]    | 276           | 396           | 414           | 750           | N/A               |
|             | Torque [ft-lb]    | 2036          | 2921          | 3053          | 5532          | N/A               |

1. 41355 option is not available for 10" API 6A 41005 valves

### A5.4 Assembly of the Bonnet

1. Check that the packing ring (B207), anti-extrusion rings (B208) and the guide bushing (B203) have been removed from the bonnet (B003).
2. Make sure that the body gasket (B015) is fitted in the Body (B001) and the Bonnet (B003) groove.
3. Position the bonnet (B003) above the valve, so that the packing flange studs (B220) are perpendicular to the flow direction of the valve.
4. Lower the bonnet (B003) onto the valve stem (B150) and push it down carefully until it goes into the valve body studs (B002) and takes its correct position.
5. Make sure that the valve plug (B112) operates freely. For the 41305 valves, take in account the friction force of the plug seal ring (B118).
6. There is no need to lubricate the stud thread as it is coated with PTFE. During the body nut tightening, take care that the plug stays in contact with the seat ring. For 41300 Series, it is necessary to apply a constant force on the plug stem (B150) to achieve and maintain this contact.
7. Screw on the body stud nuts (B014) by hand. Tighten the nuts lightly and evenly so that the internal parts are held in place. The face of the bonnet (B003) should be parallel to the upper face of the body (B001).
8. Slide the guide bushing (B203) onto the valve plug stem (B150) and let it drop to the bottom of the packing box housing.

### A5.5 Tightening of Body Studs

In order to achieve alignment of the cage (B118) and the valve plug (B112), force must be applied to the plug stem (B150) during tightening of the bonnet (B003) which results in correct relative positioning of the two parts.

The force can be applied with the pneumatic actuator as follows:

Place the actuator on the valve bonnet (B003) by means of the yoke nut (B017) or attachment screws (B915) and connect the valve plug stem (B150) to the actuator stem.

# CAUTION

During this operation, make sure that the plug does not turn when it is seated. If the plug travel is very small and there is a large amount of plug stem inside the actuator, it may be necessary to remove the yoke nut and lift the actuator so that the plug is not touching the seat.

Align the internal parts as follows:

1. In the case of air-to-extend actuators, supply air to the actuator at the maximum pressure indicated on the serial plate and in the case of spring-to-extend actuators, do not supply air to the actuator, so that the optimum positioning of the valve plug (B112) and cage (B107) can be obtained.
2. Single bolting circle: in all the case, mount the nuts in order to indicate by Figures 24 and 25 on page 36 and screw them by successive, uniform and progressive levels. Make sure that the bonnet's face stays parallel at the body one, up to the torque value given in Tables 8 and 9 on page 23.

# ! WARNING

Due to the use of metal body gasket(B015), the body (B001) and the bonnet (B003) will NEVER be in metal/metal contact.

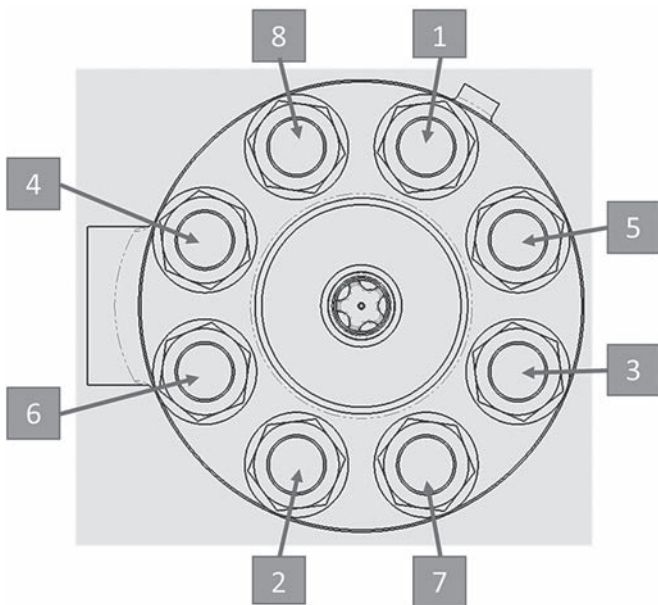


Figure 24 - Torque Sequence Single Bolting Circle 8 Bolts

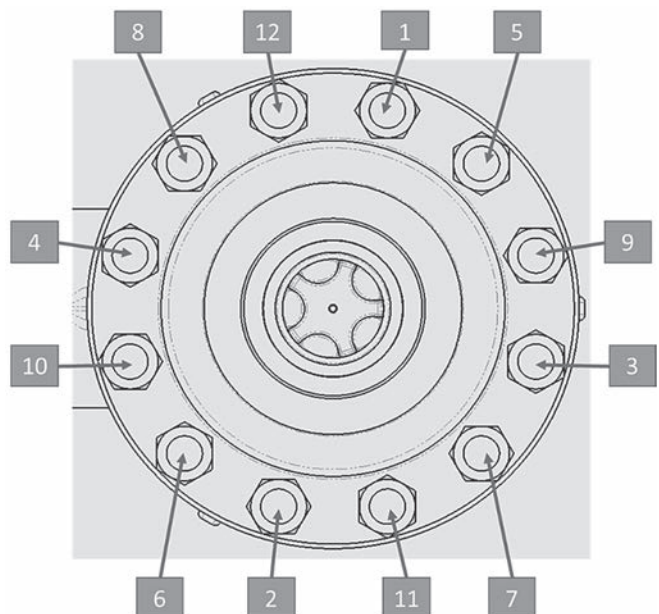


Figure 25 - Torque Sequence Single Bolting Circle 12 Bolts



**Table 14 - Torque Requirements Body/ Bonnet 10K Valves**

|                        | Valve Size (inch) | 3            | 4            | 6            | 8            | 10        |
|------------------------|-------------------|--------------|--------------|--------------|--------------|-----------|
|                        | Valve Class       | 10K          | 10K          | 10K          | 10K          | 10K       |
| Stud Material          | Bolt Thread size  | 1"3/4-8UN-2A | 1"1/2-8UN-2A | 1"3/4-8UN-2A | 1"7/8-8UN-2A | 2"-8UN-2A |
|                        | Bolt quantity     | 8            | 8            | 8            | 12           | 12        |
| Body studs<br>B7M /L7M | Torque [daN.m]    | 150          | 90           | 149          | 185          | 225       |
|                        | Torque [ft-lb]    | 1106         | 664          | 1099         | 1364         | 1660      |
|                        | Max Torque[daN.M] | 164          | 101          | 164          | 202          | 250       |
|                        | Max Torque[ft-lb] | 1210         | 745          | 1210         | 1490         | 1844      |
| Body studs<br>B7/ L7   | Torque [daN.m]    | 195          | 120          | 195          | 240          | 295       |
|                        | Torque [ft-lb]    | 1438         | 885          | 1438         | 1770         | 2176      |
|                        | Max Torque[daN.M] | 215          | 132          | 215          | 266          | 325       |
|                        | Max Torque[ft-lb] | 1586         | 974          | 1586         | 1962         | 2397      |

**Table 15 - Torque Requirements Body/ Bonnet 15K Valves**

| Description            | Valve Size (inch) | 3            | 4            | 6            | 8            | 10           |
|------------------------|-------------------|--------------|--------------|--------------|--------------|--------------|
|                        | Valve Class       | 15k          | 15k          | 15k          | 15k          | 15k          |
| Stud Material          | Bolt Thread size  | 1"7/8-8UN-2A | 1"3/4-8UN-2A | 2"1/4-8UN-2A | 2"1/4-8UN-2A | 2"1/2-8UN-2A |
|                        | Bolt number       | 8            | 8            | 8            | 12           | 12           |
| Body studs<br>B7M /L7M | Torque [daN.m]    | 185          | 150          | 226          | 330          | 450          |
|                        | Torque [ft-lb]    | 1364         | 1106         | 1667         | 2434         | 3319         |
|                        | Max Torque[daN.M] | 202          | 164          | 250          | 360          | 500          |
|                        | Max Torque[ft-lb] | 1490         | 1210         | 1844         | 2655         | 3688         |
| Body studs<br>B7/ L7   | Torque [daN.m]    | 240          | 195          | 296          | 430          | 595          |
|                        | Torque [ft-lb]    | 1770         | 1438         | 2183         | 3172         | 4388         |
|                        | Max Torque[daN.M] | 266          | 215          | 326          | 470          | 650          |
|                        | Max Torque[ft-lb] | 1962         | 1586         | 2404         | 3467         | 4794         |

## A6. API 6A Body Sub-Assembly Cross Section

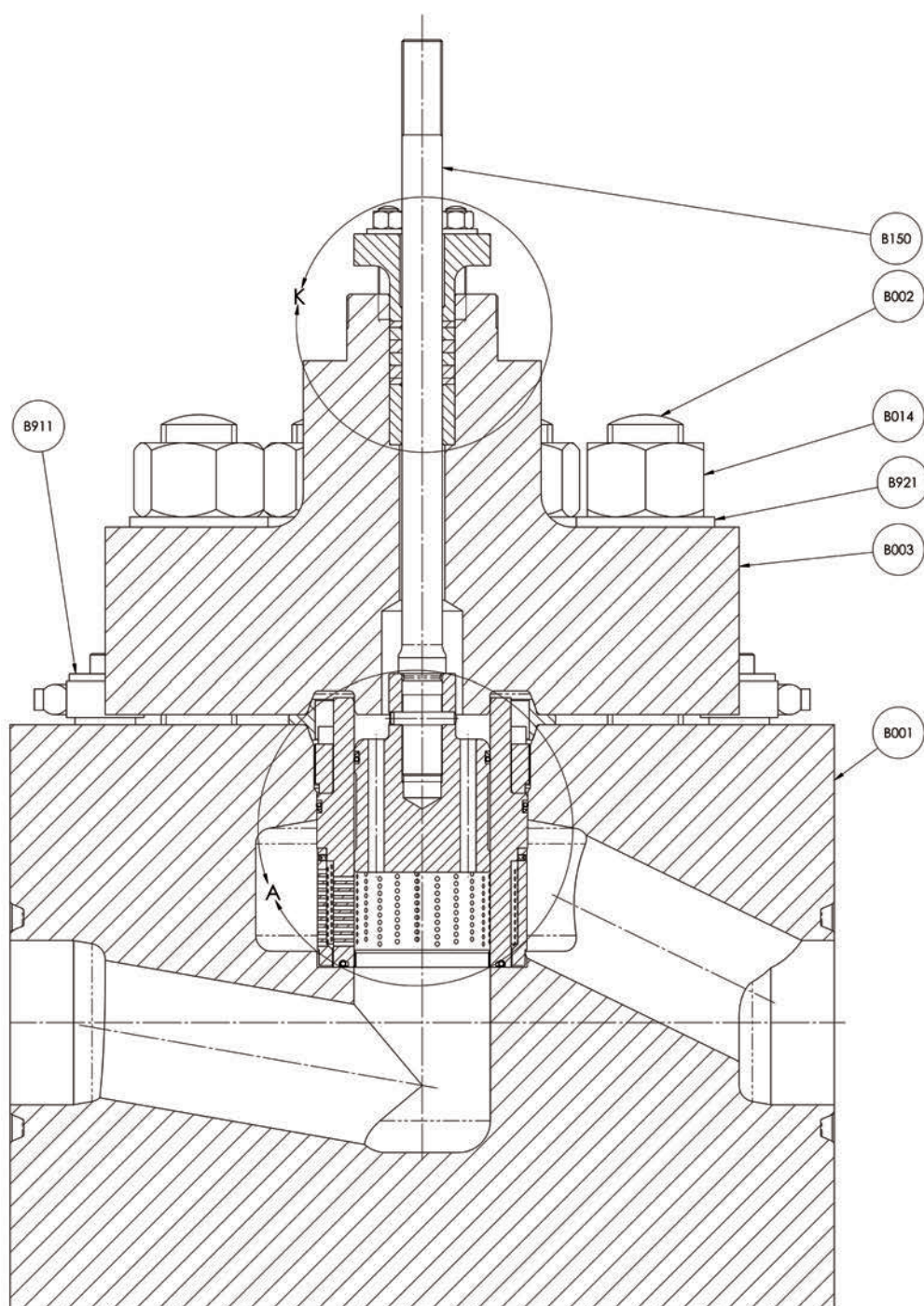


Figure 26 - Cross Sectional View of 41005 API 6A Design

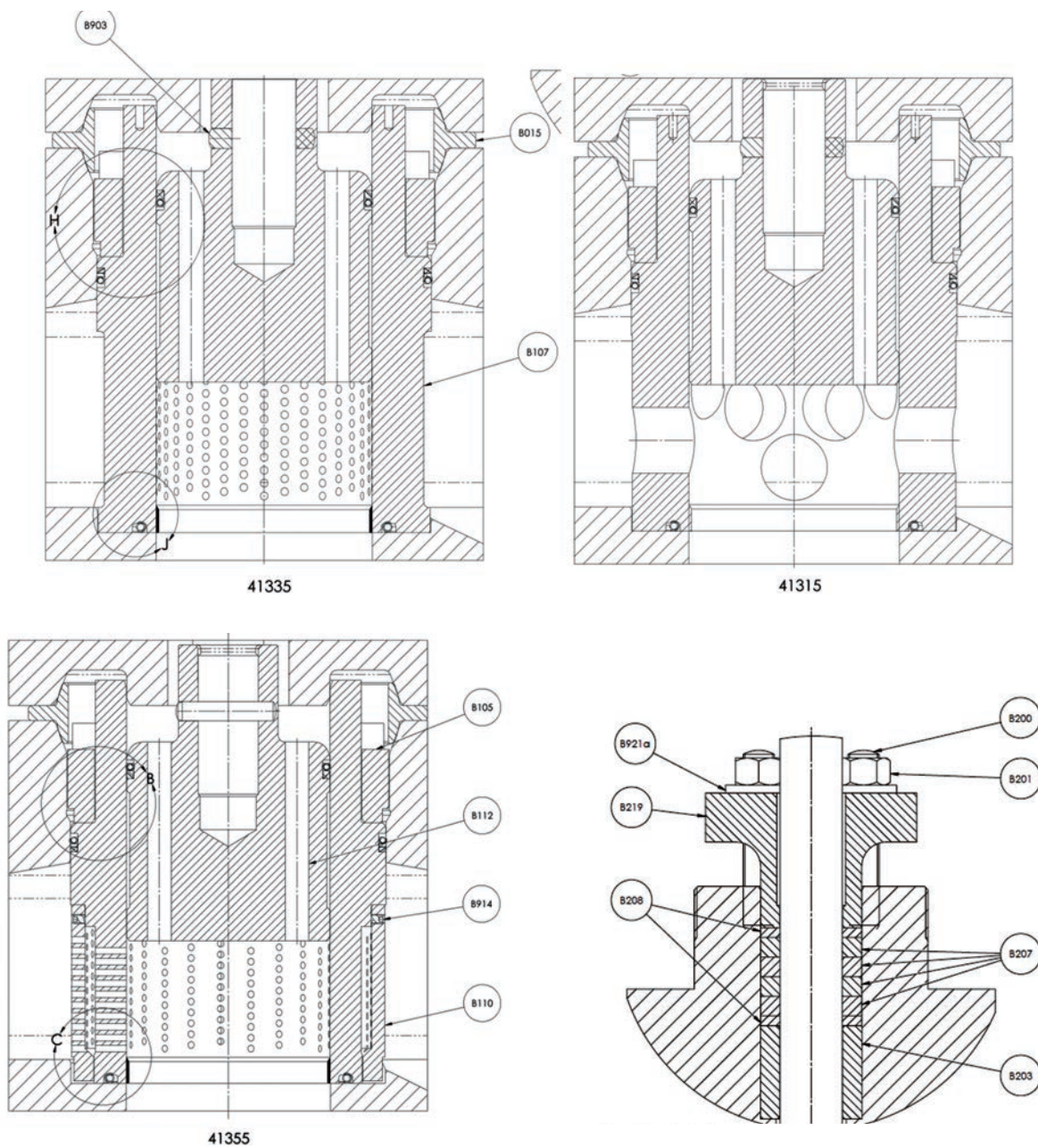


Figure 27 - Trim Type and Packing Detail

**Table 16 - Body Sub-Assembly Parts Reference and Spare Parts Code**

| Ref. No. | Spare Part Code | Part Description     |
|----------|-----------------|----------------------|
| B001     | N               | Valve body           |
| B002     | N               | Body stud            |
| B003     | N               | Valve bonnet         |
| B014     | N               | Body nuts            |
| B015     | C               | Body gasket          |
| B017     | N               | Drive nut, not shown |
| B103     | C               | Seat ring gasket     |
| B105     | N               | Trim retainer        |
| B107     | W               | Cage                 |
| B108     | C               | Cage gasket          |
| B110     | W               | Outer cage           |
| B112     | W               | Plug                 |
| B118     | C               | Plug balanced seal   |

| Rev. No. | Spare Part Code | Part Description                  |
|----------|-----------------|-----------------------------------|
| B150     | W               | Plug stem                         |
| B203     | W               | Packing bushing                   |
| B207     | C               | Packing rings                     |
| B208     | C               | Anti-extrusion rings              |
| B219     | N               | Packing flange follower           |
| B220     | N               | Packing stud                      |
| B221     | N               | Packing nut                       |
| B903     | W               | Plug pin                          |
| B911     | N               | Lifting lug                       |
| B915     | N               | Yoke and bonnet screws, not shown |
| B914     | C               | Cage set screw                    |
| B921     | N               | Body plain washer                 |
| B921a    | N               | Packing plain washer              |

Notes:

C = Consumable

W = Wear Parts

N = Not a Spare Part

## A7. Actuation

Refer to the instruction, operation and maintenance manuals for the 51/52/53 Series Pneumatic Piston Actuators and the 87/88 Series Actuators.

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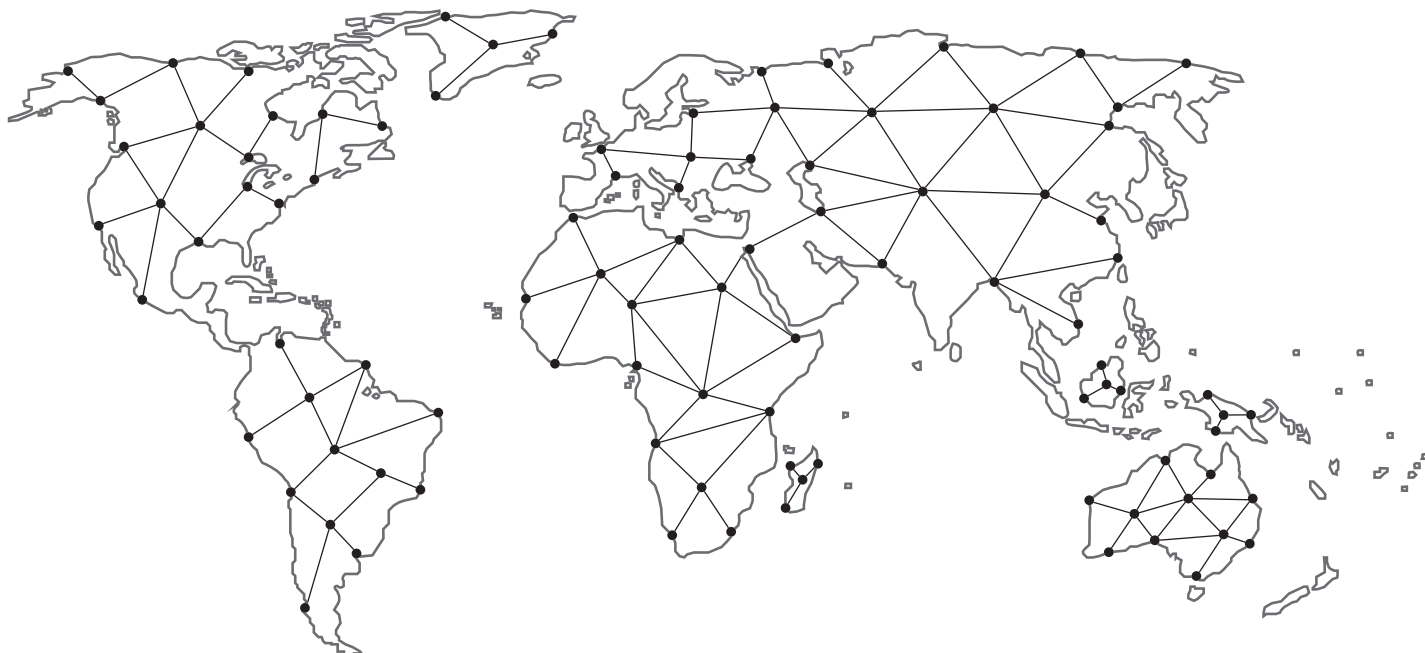
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